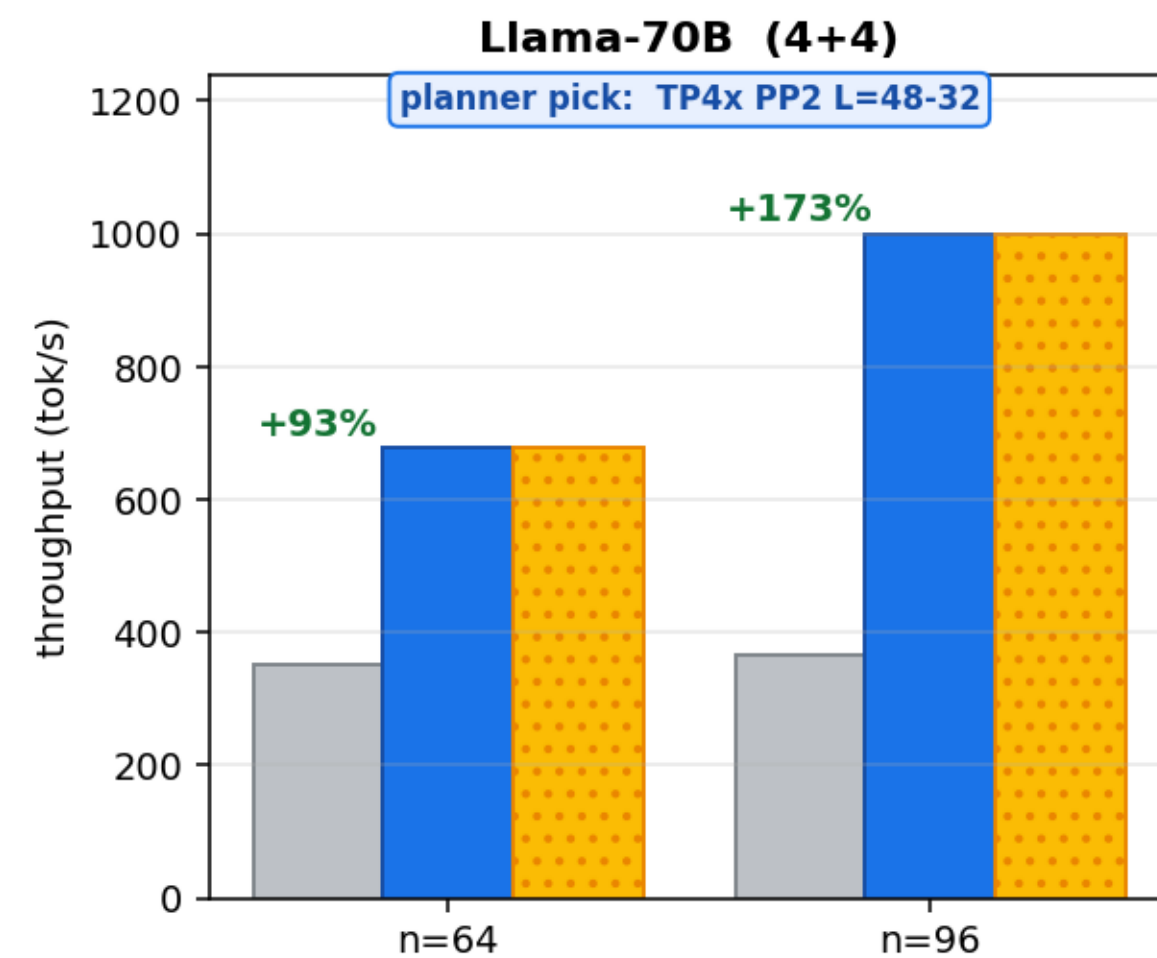
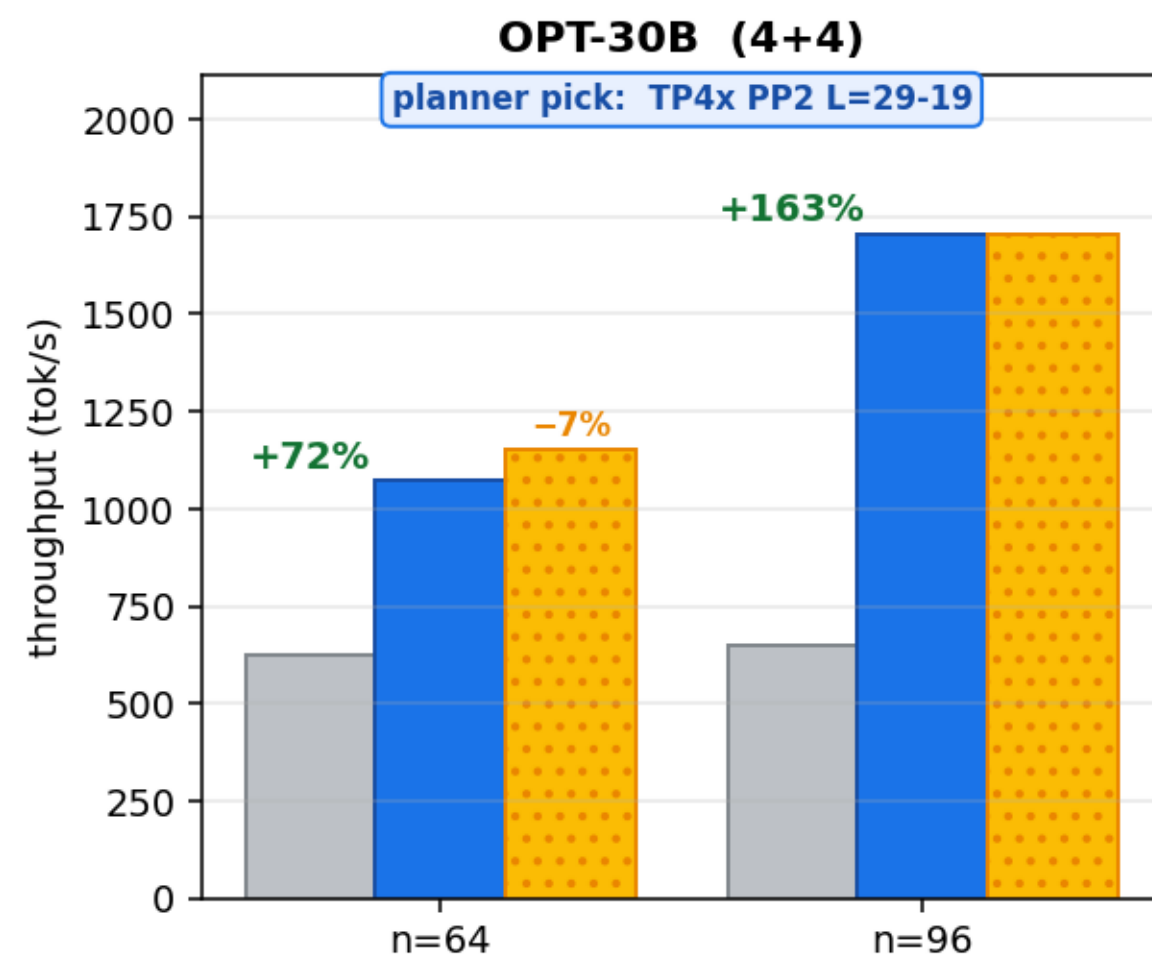
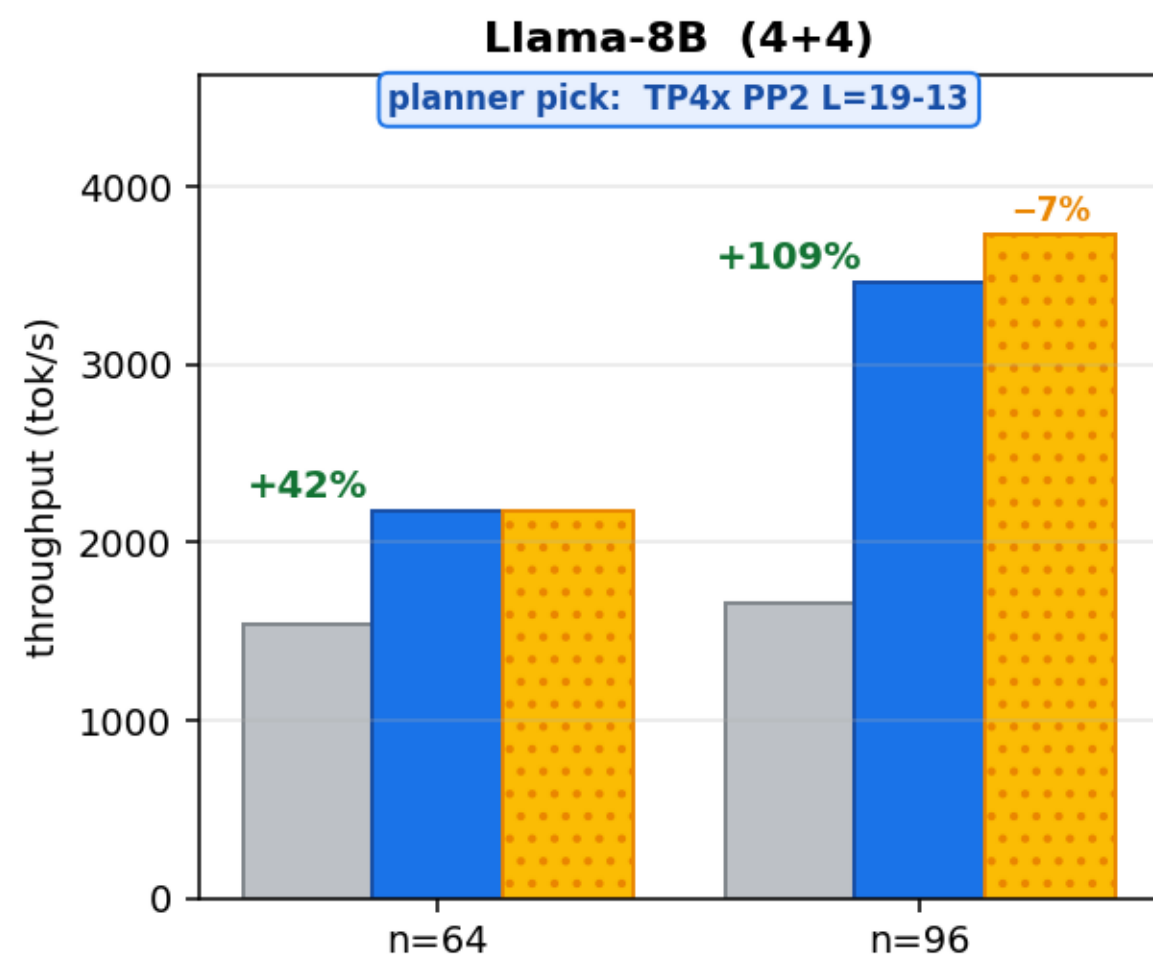
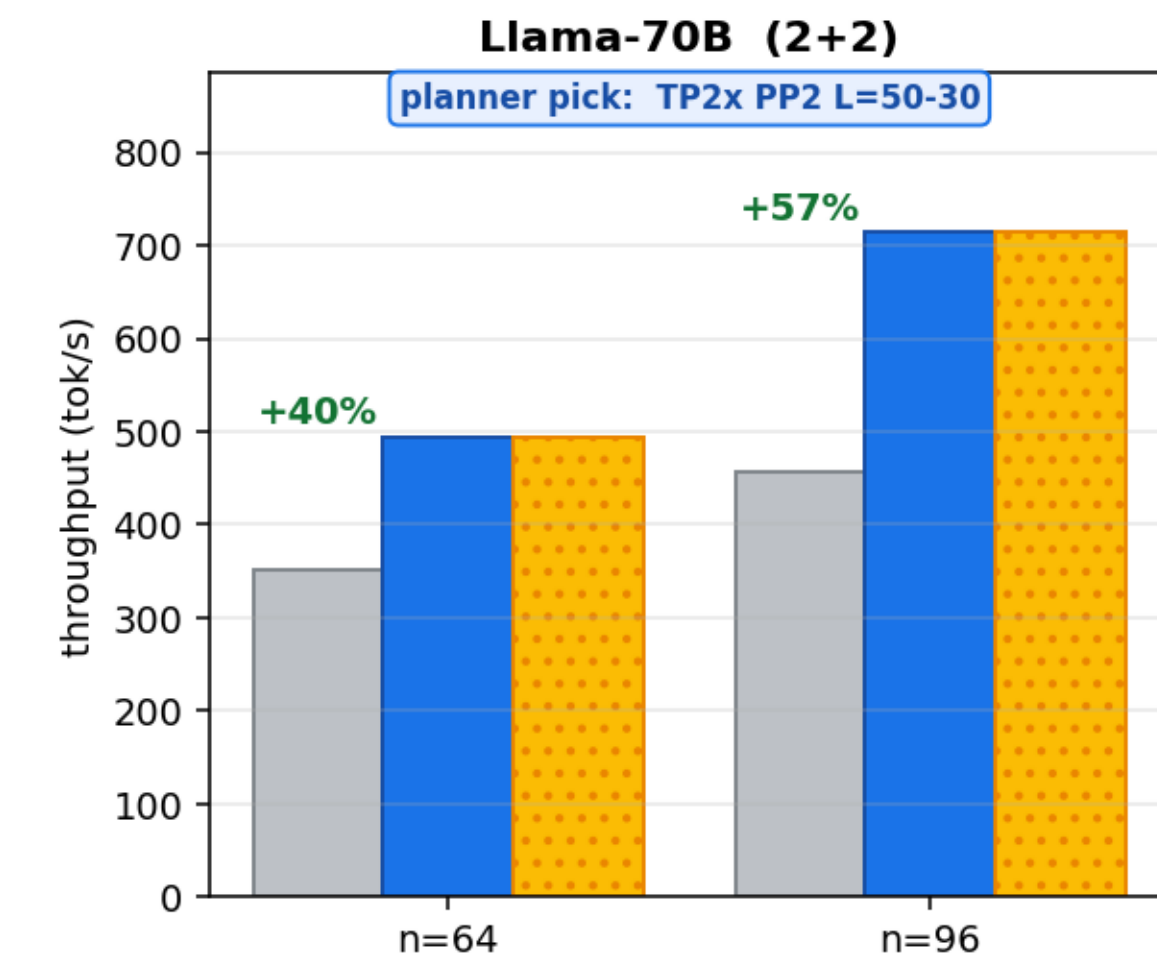
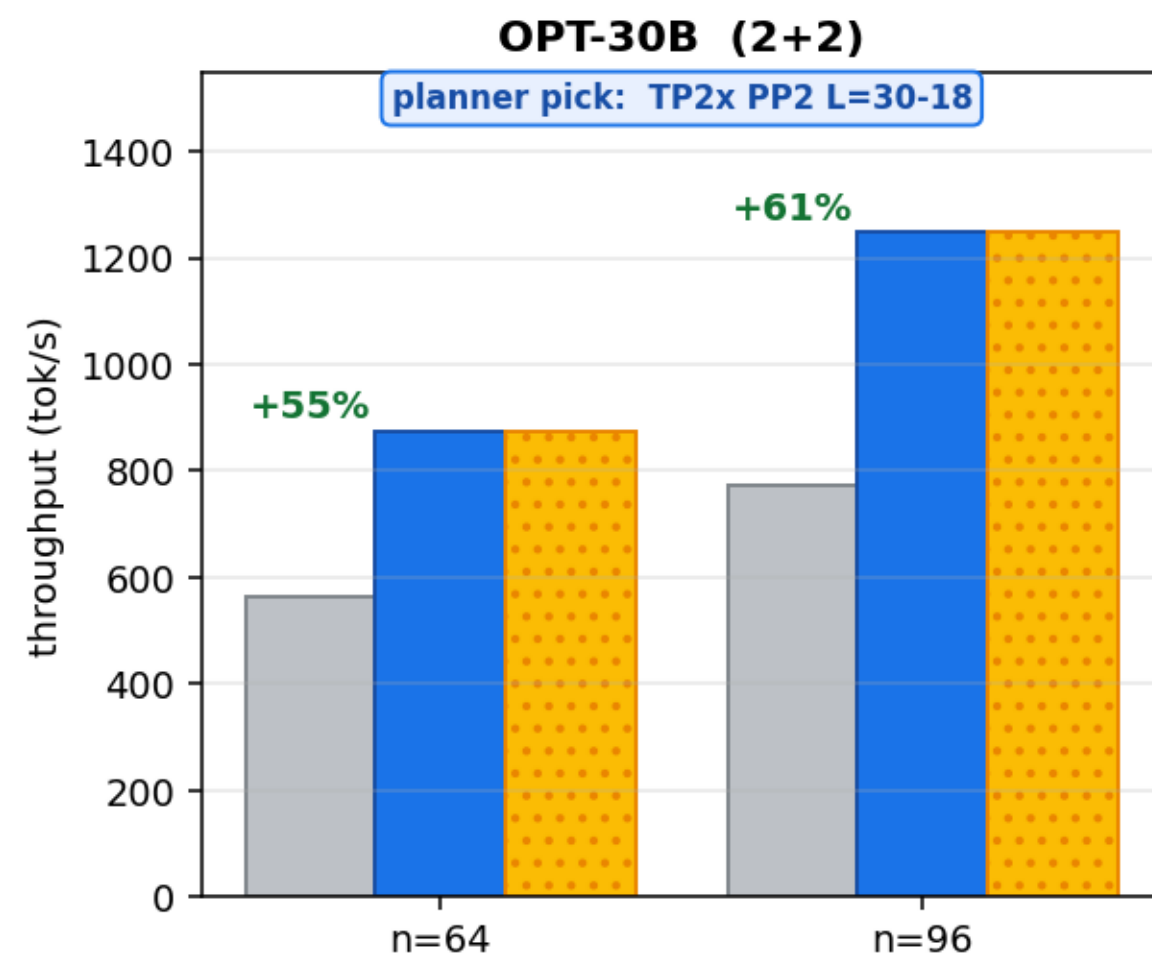
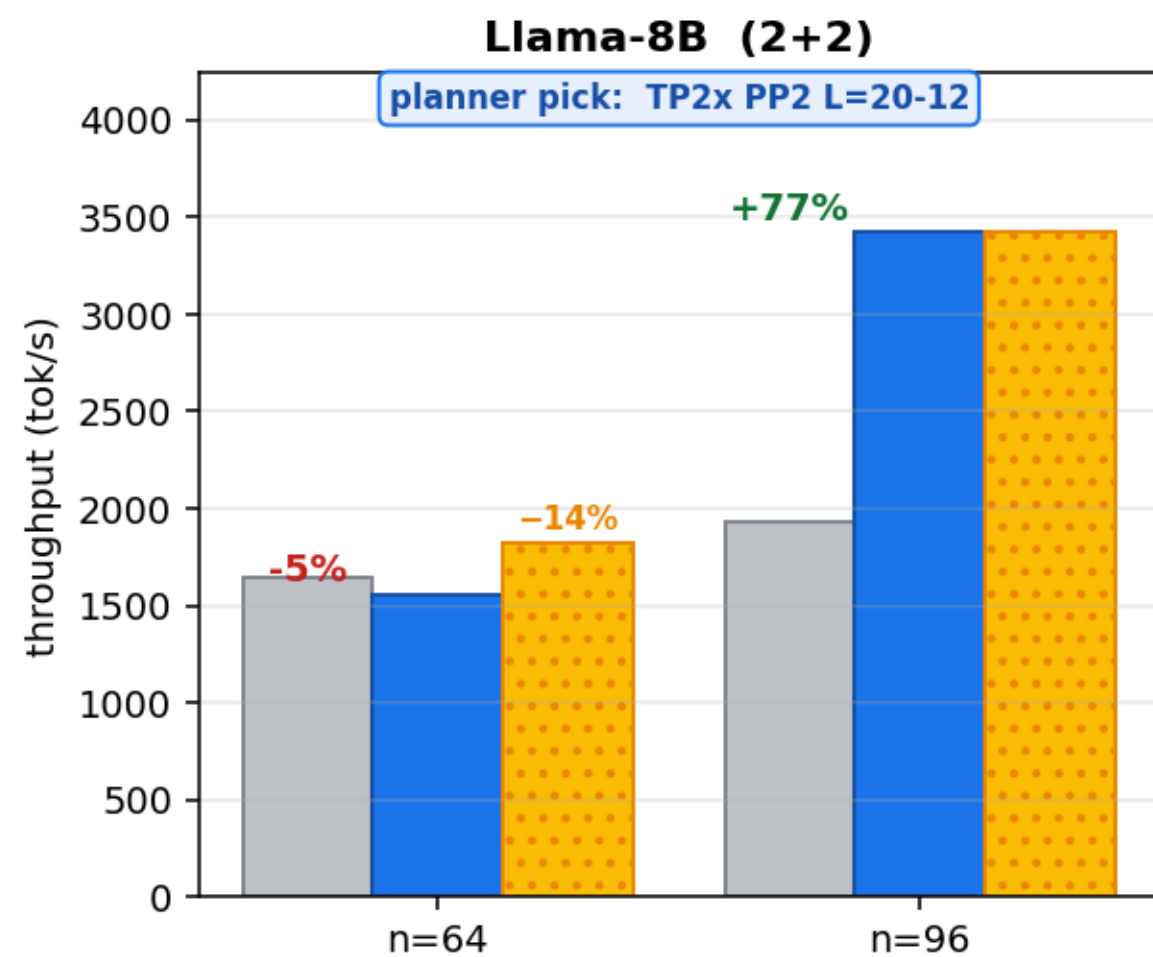


Mixed-traffic serving — planner pick vs uniform baseline vs measured oracle
(varied (input,output) shapes per request in one concurrent stream)

4+4
layout

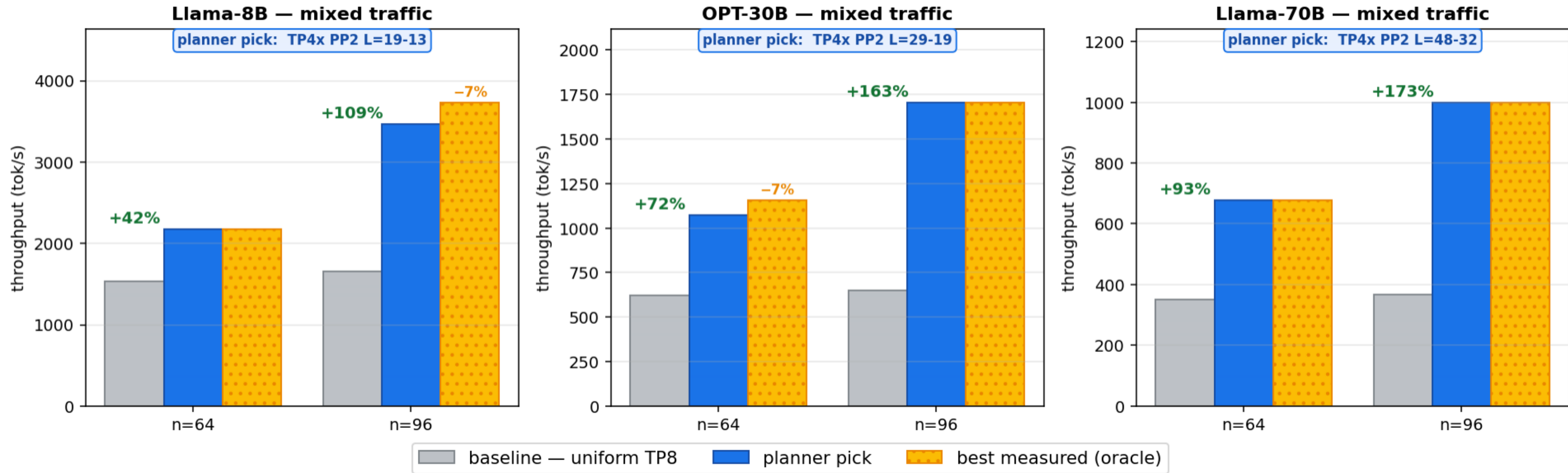


2+2
layout



baseline — uniform TP=world (per layout) planner pick best measured (oracle)

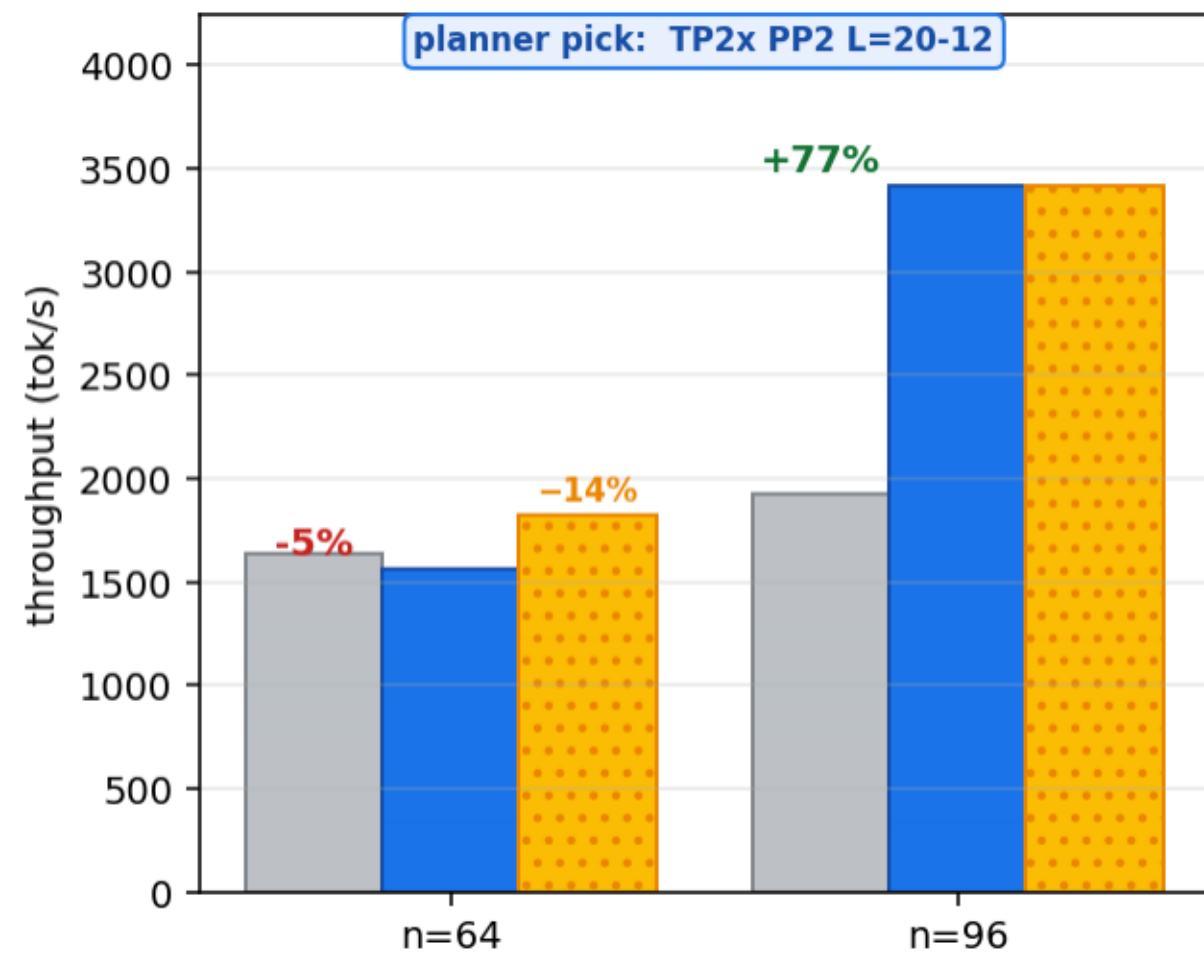
Mixed-traffic serving: varied (input,output) shapes per request in one stream — planner pick vs uniform baseline (4+4)



Mixed-traffic serving: varied (input,output) shapes per request in one stream — planner pick vs uniform baseline (2+2)

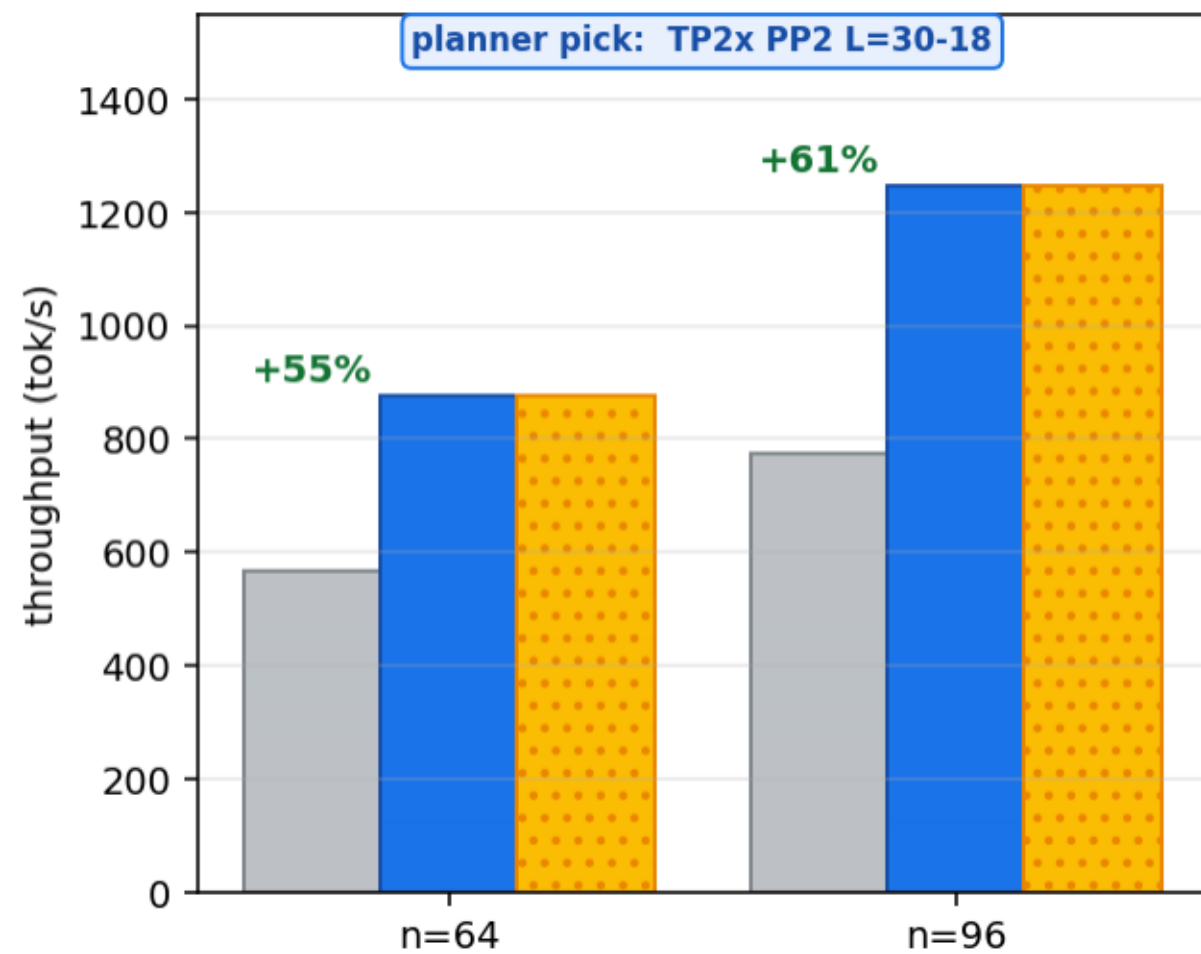
Llama-8B — mixed traffic

planner pick: TP2x PP2 L=20-12



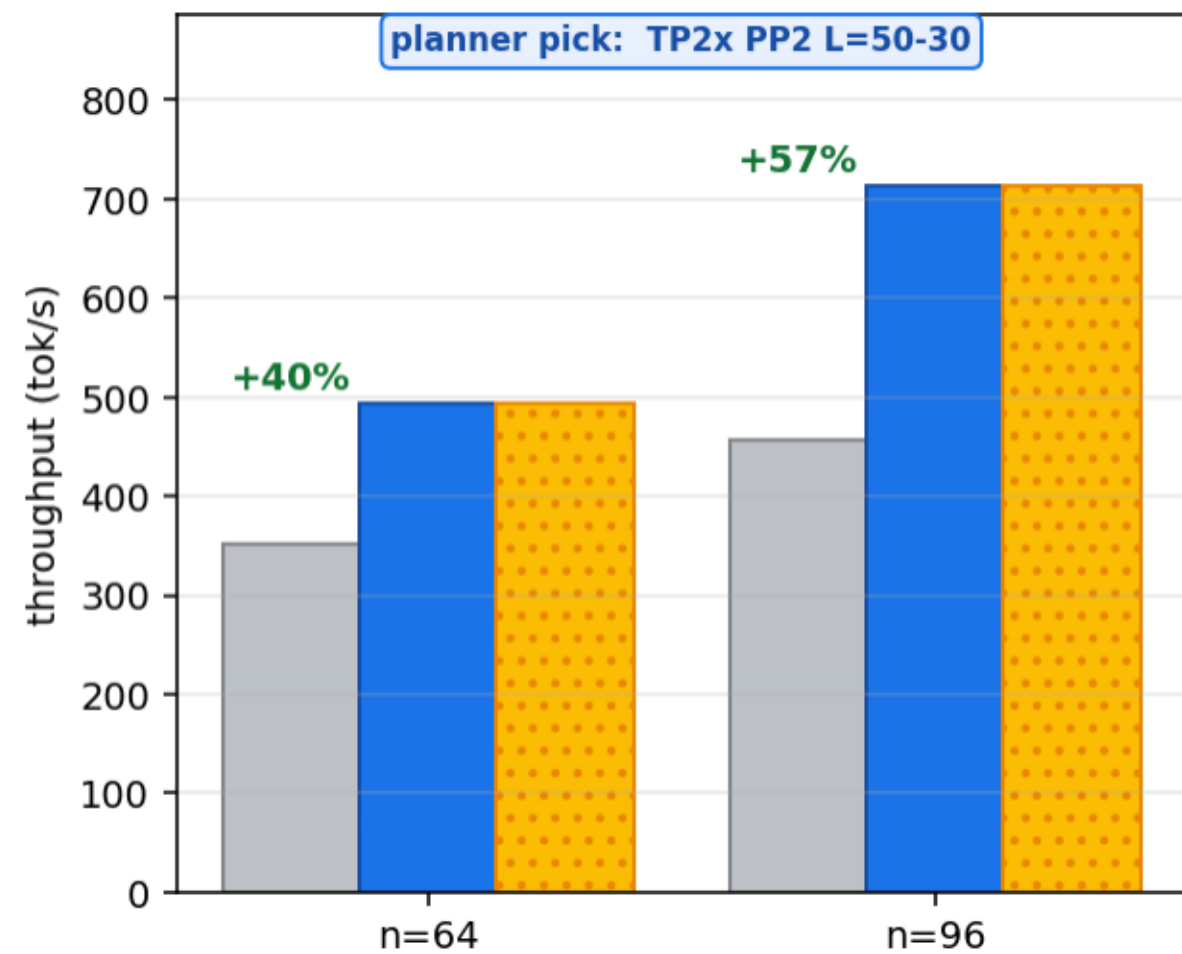
OPT-30B — mixed traffic

planner pick: TP2x PP2 L=30-18



Llama-70B — mixed traffic

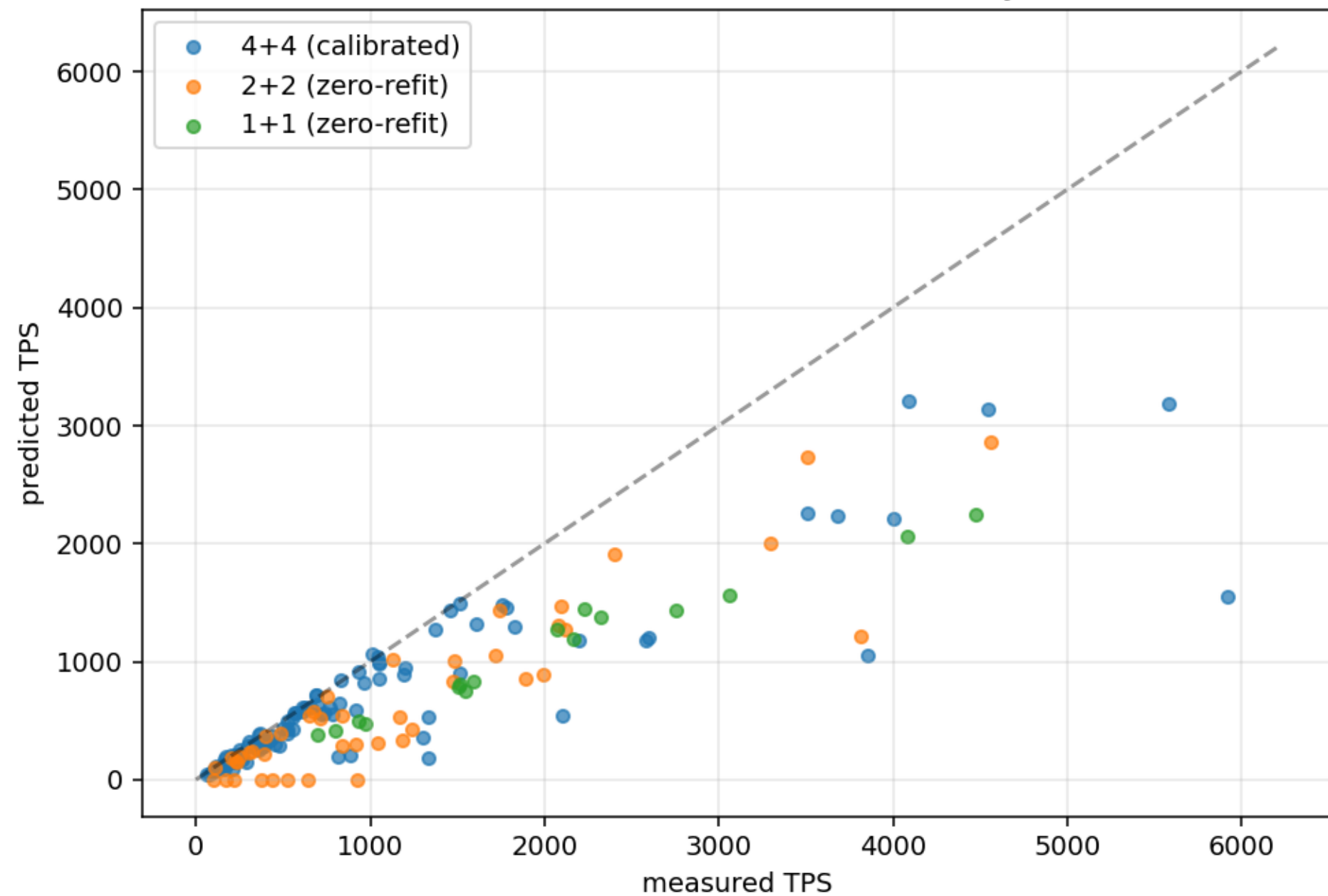
planner pick: TP2x PP2 L=50-30



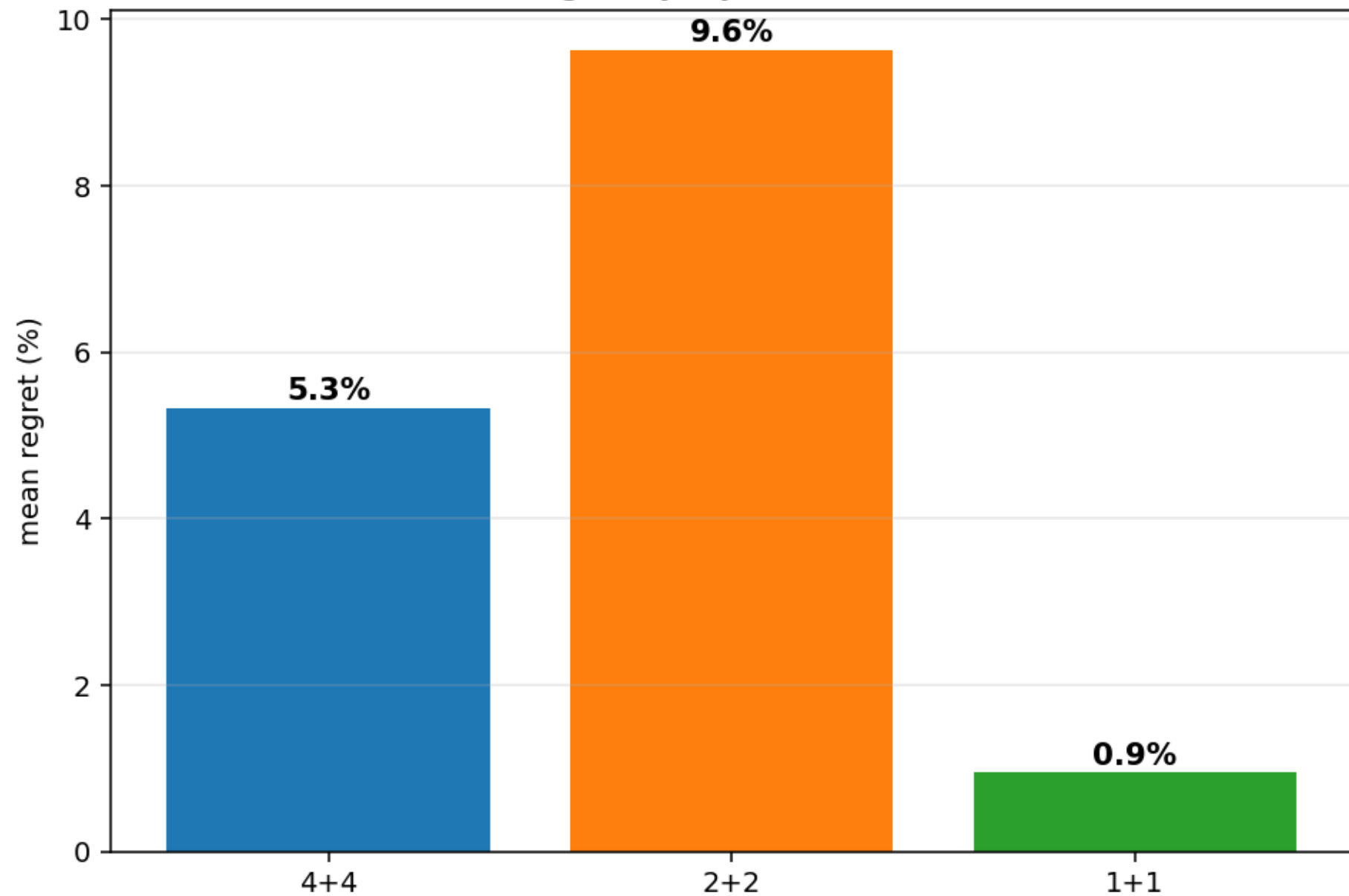
baseline — uniform TP4 planner pick best measured (oracle)

Planner accuracy: calibrated on 4+4, generalizes zero-refit to 2+2 / 1+1

Predicted vs measured (balanced, all layouts)



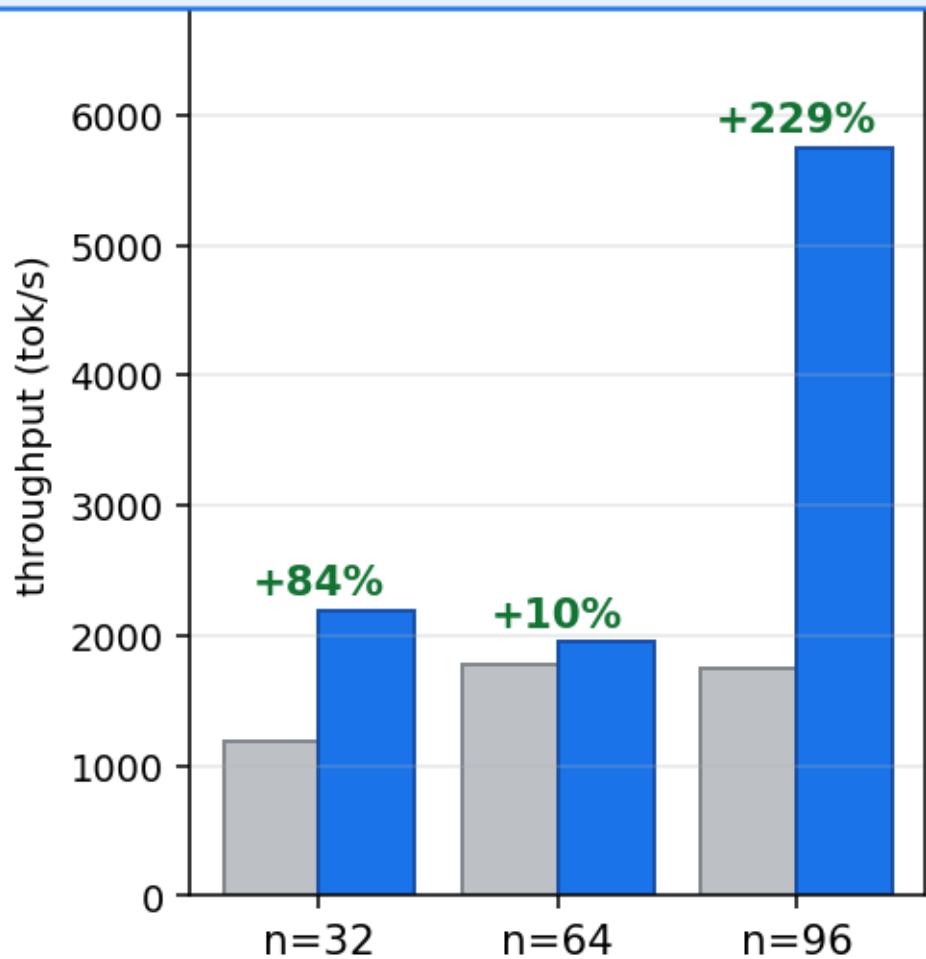
Planner regret by layout (lower = better)



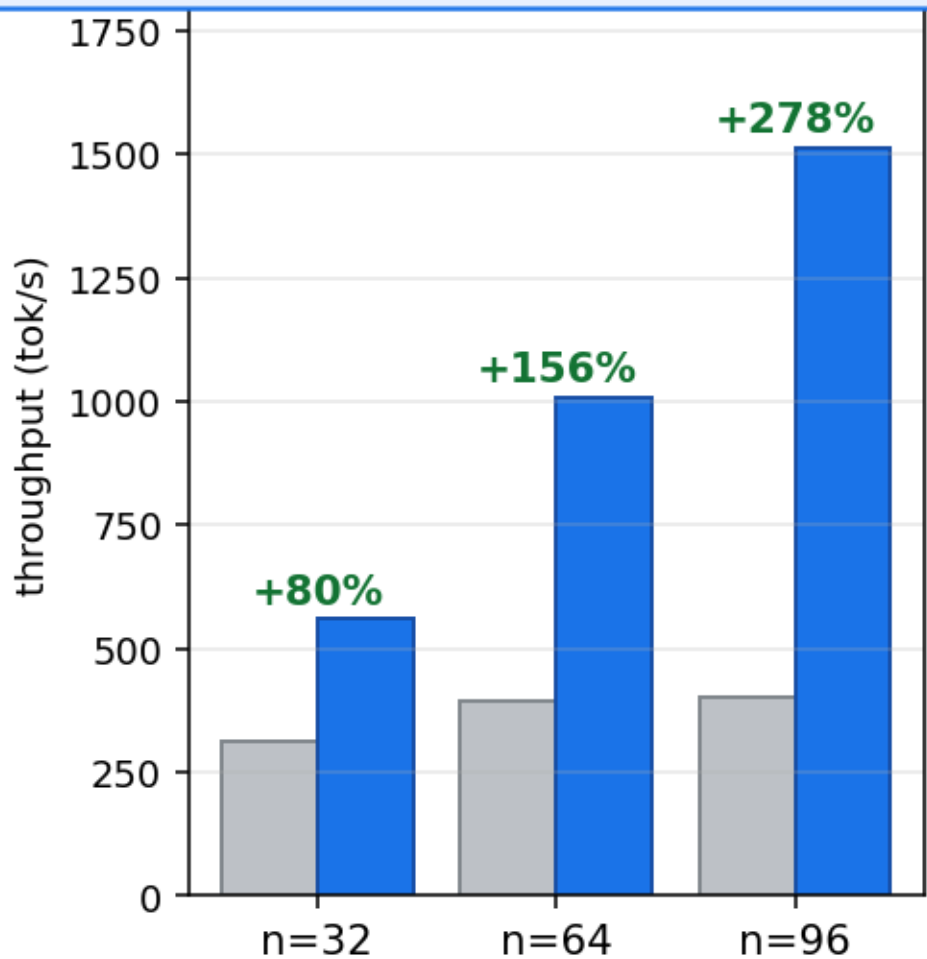
Self-validation on a HELD-OUT workload (chat in=768/out=256, not in calibration): raw planner pick vs naive baseline
raw predicted-best config (no safety guard) — wins big at production load

Planner (raw top-1 pick) vs naive baseline (uniform TP8) — balanced, 4+4, n>=32 (qwen3-32B excluded — fork PP-overlap gap, §8)

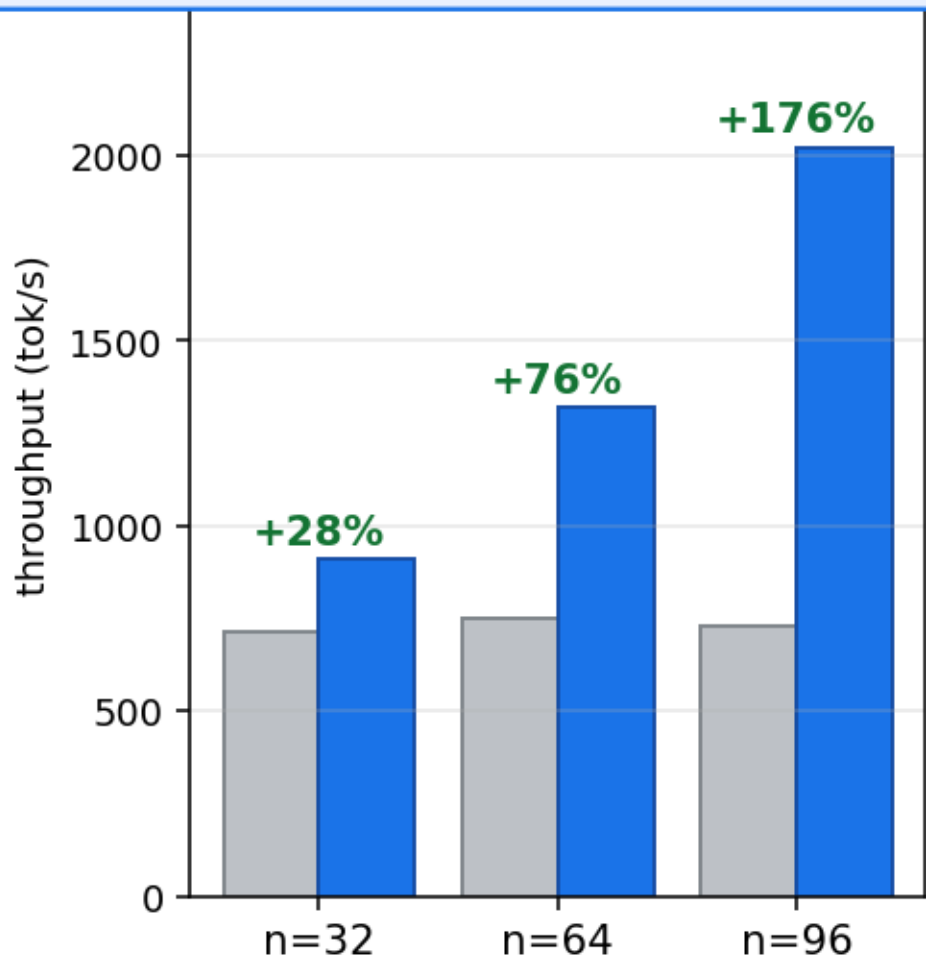
Llama-3.1-8B



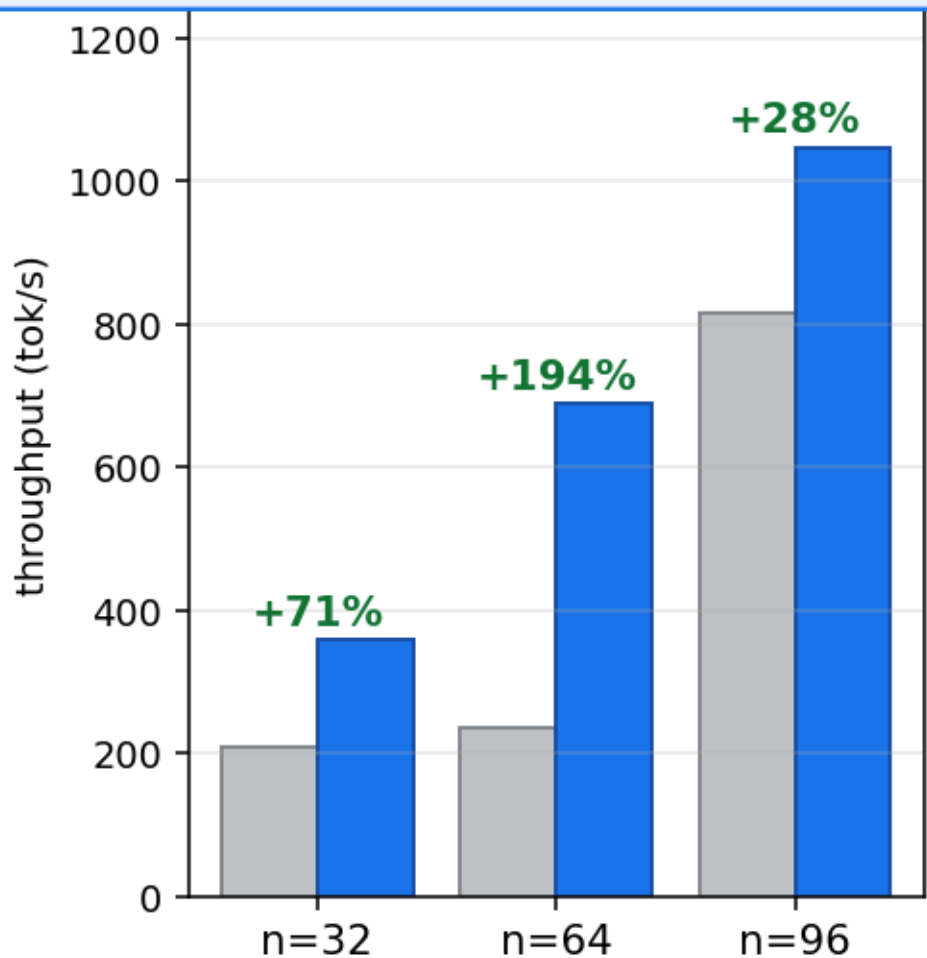
Llama-3.3-70B



OPT-30B



Mistral-Large-123B

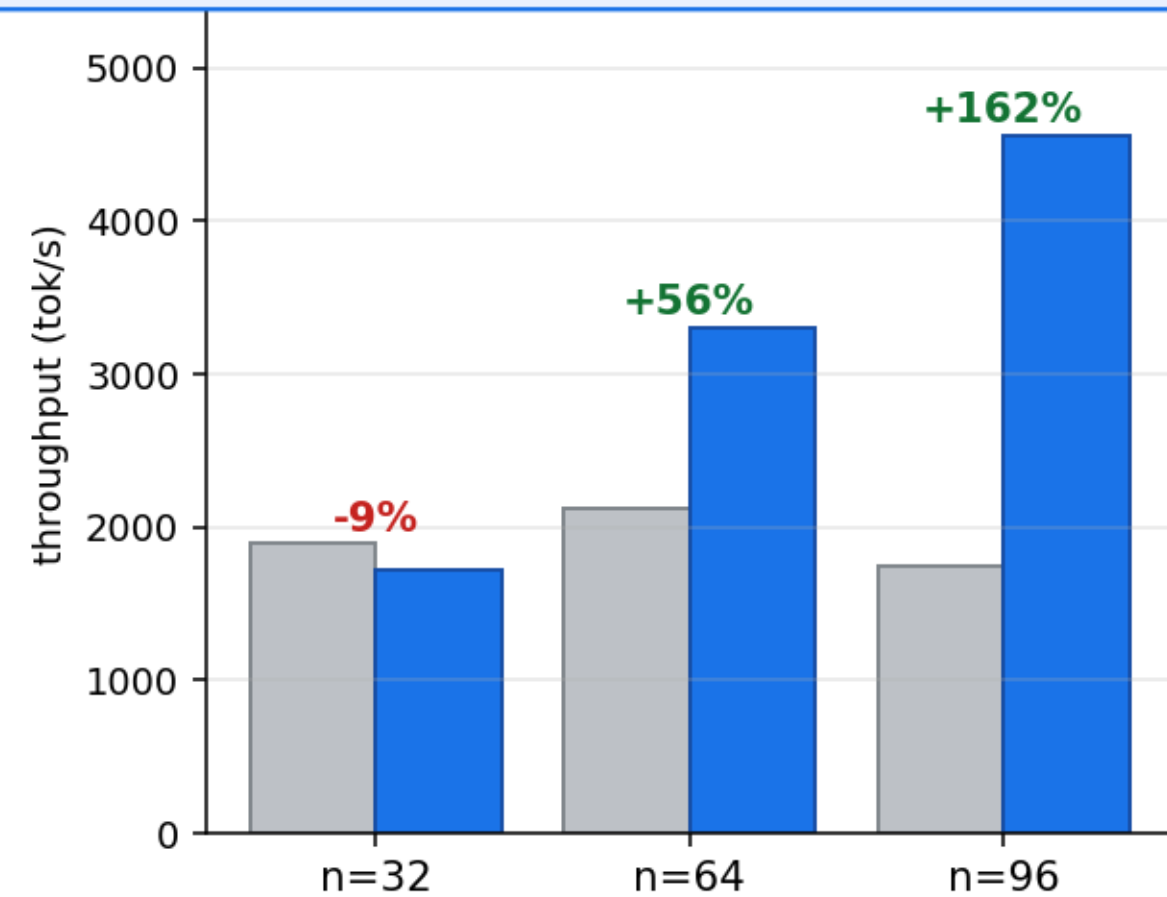


baseline — uniform TP8 planner pick (config in box above each panel)

Planner (raw top-1 pick) vs naive baseline (uniform TP4) — balanced, 2+2, $n \geq 32$ (qwen3-32B excluded — fork PP-overlap gap, §8)

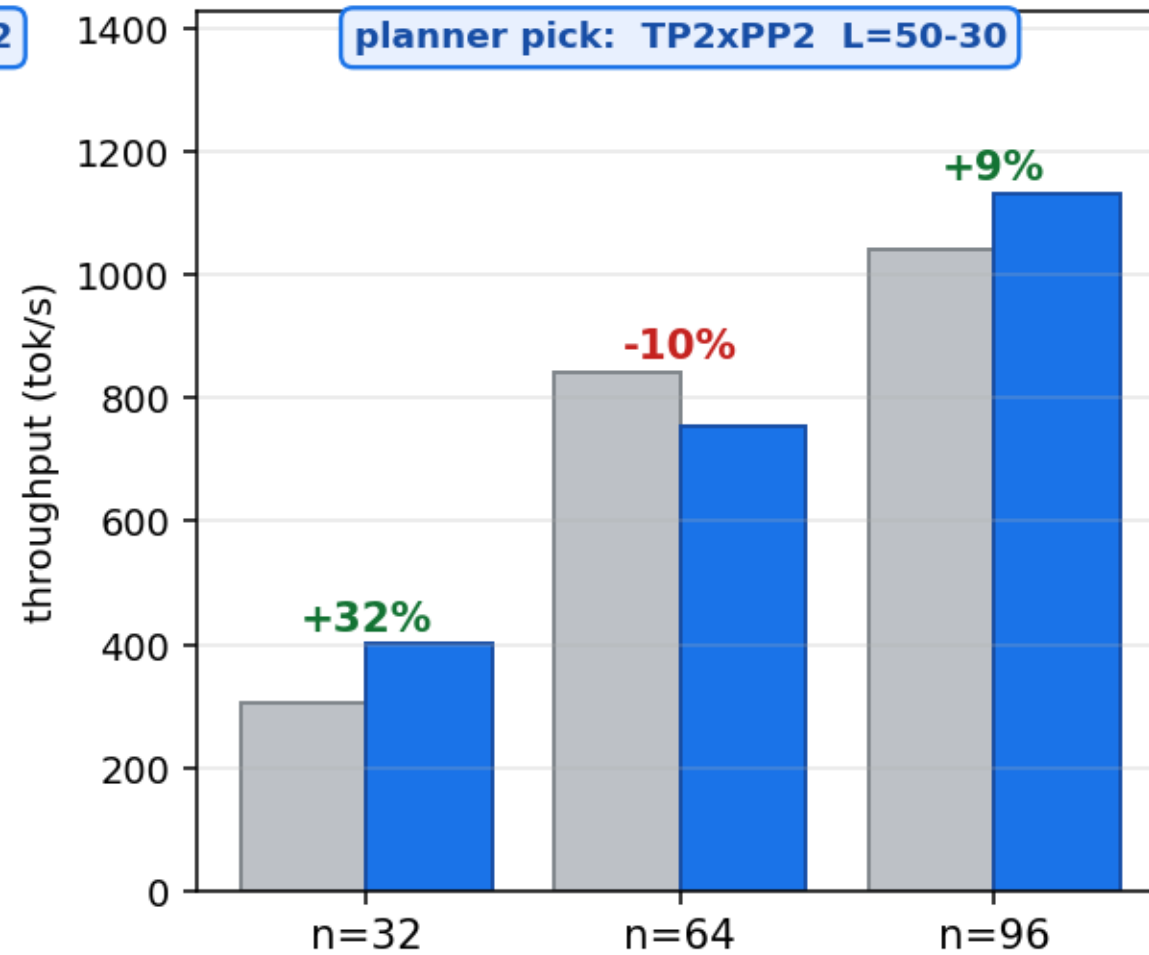
Llama-3.1-8B

planner pick: n32: TP2xPP2 L=21-11 | n64: TP2xPP2 L=20-12 | n96: TP2xPP2 L=20-12



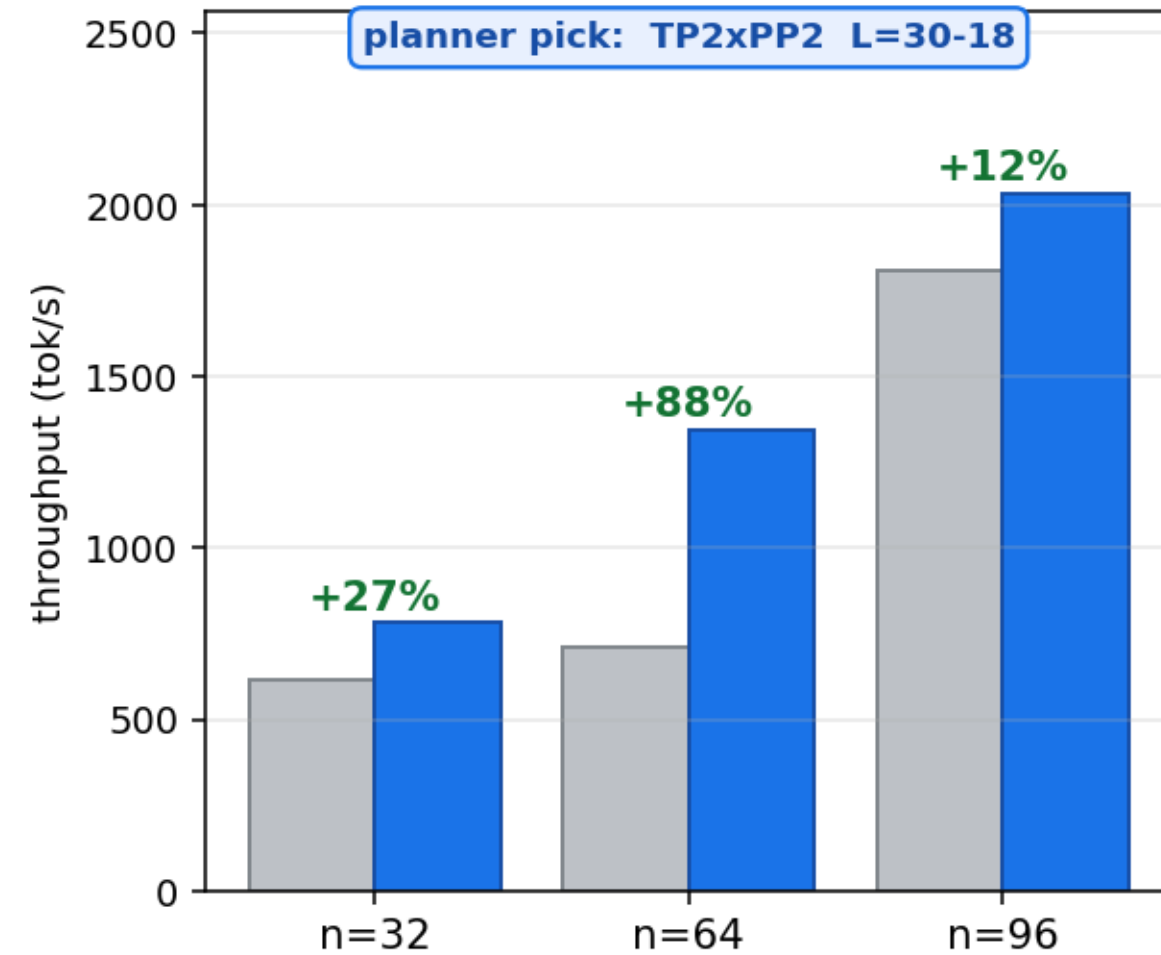
Llama-3.3-70B

planner pick: TP2xPP2 L=50-30



OPT-30B

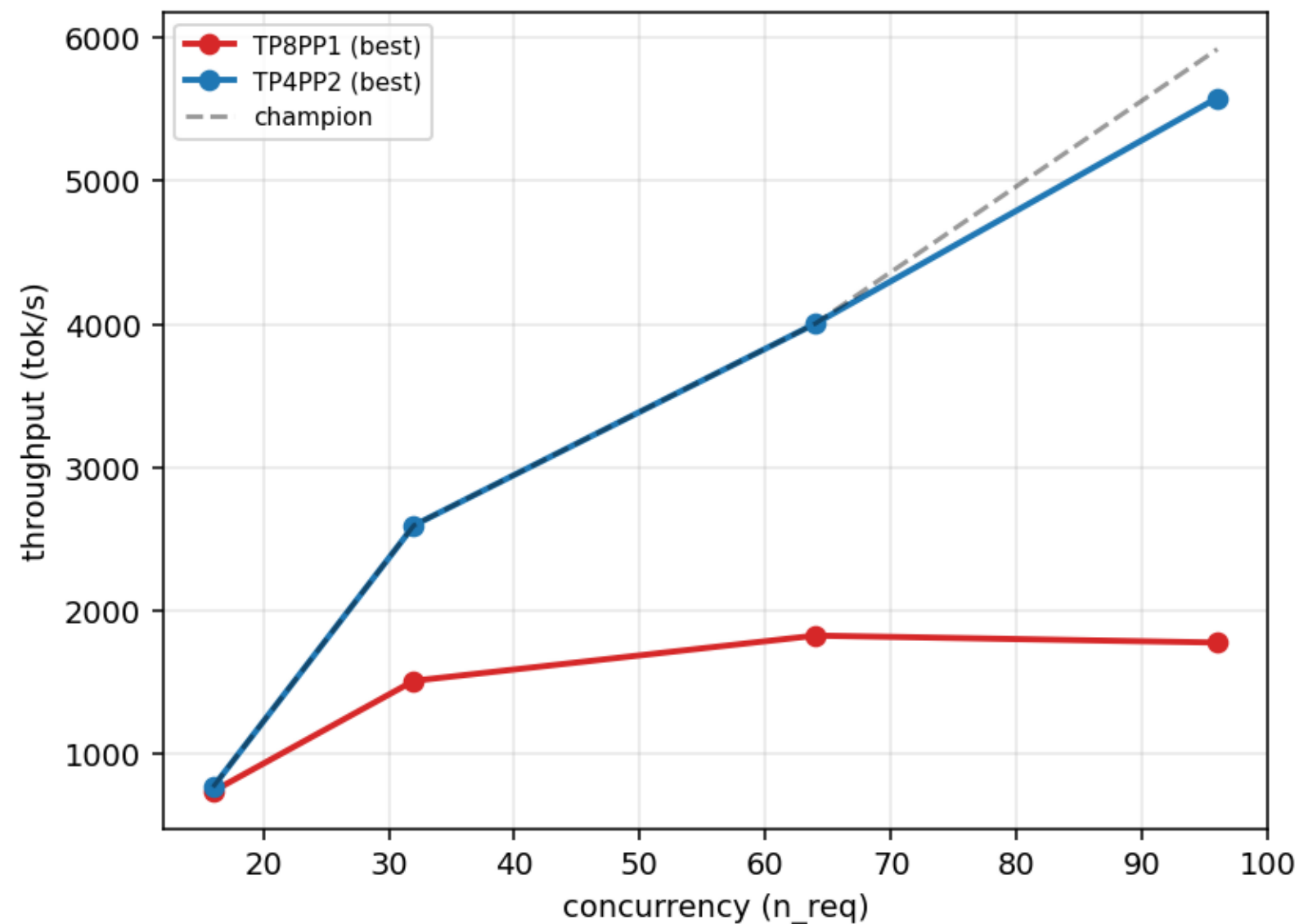
planner pick: TP2xPP2 L=30-18



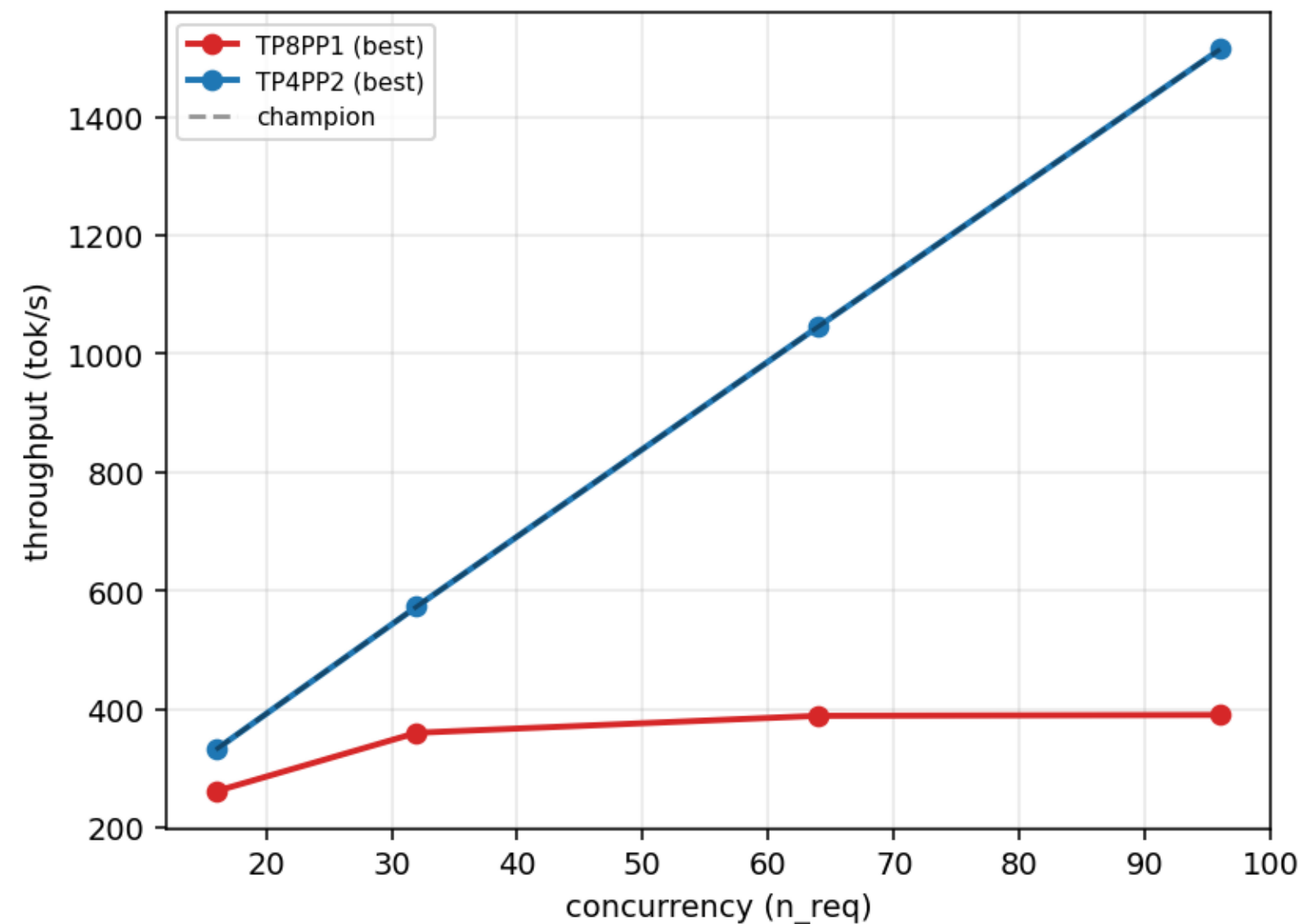
baseline — uniform TP4 planner pick (config in box above each panel)

Load-dependent topology crossover (4+4, balanced): TP-heavy at low load → PP at high load

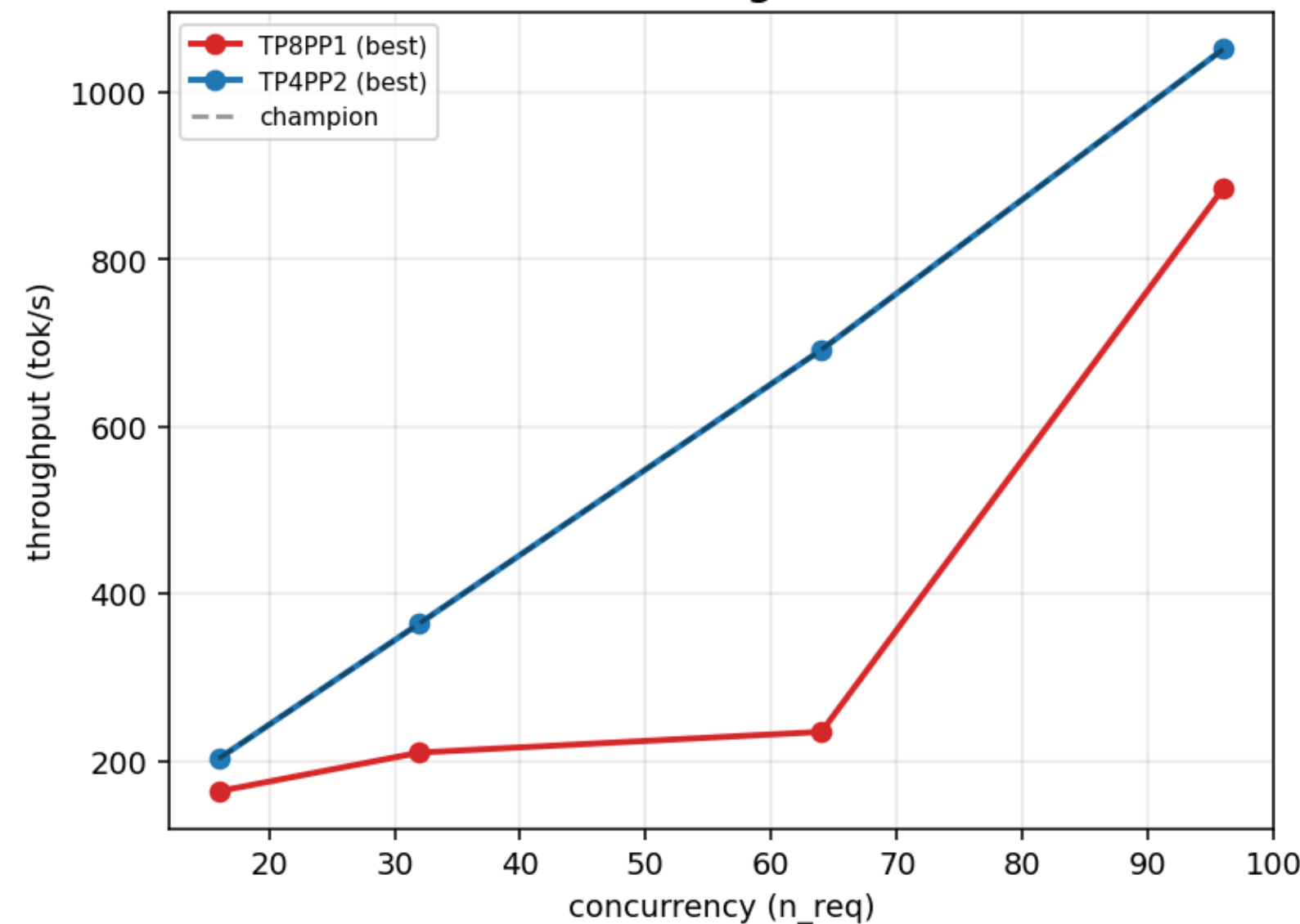
Llama-3.1-8B



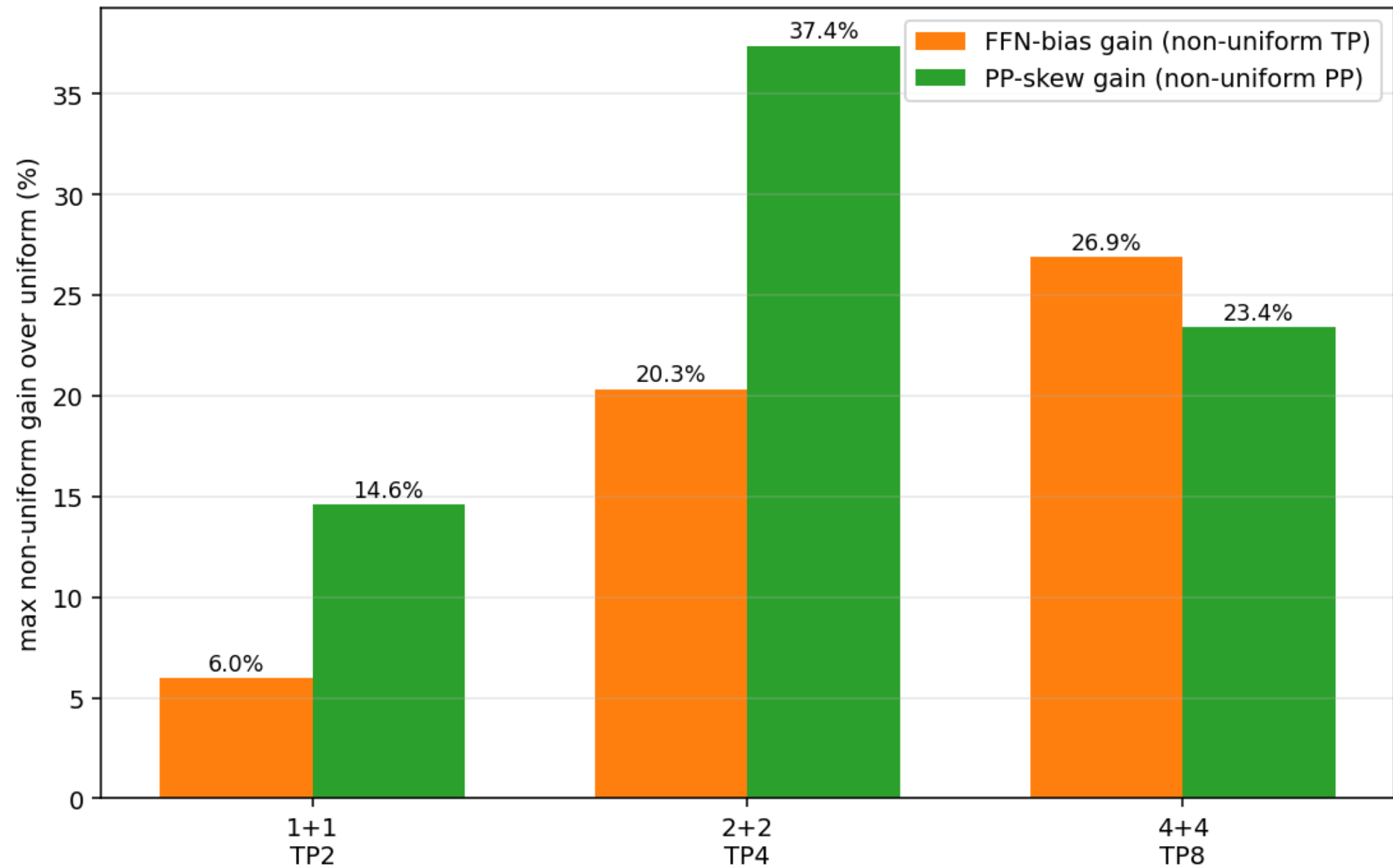
Llama-3.3-70B



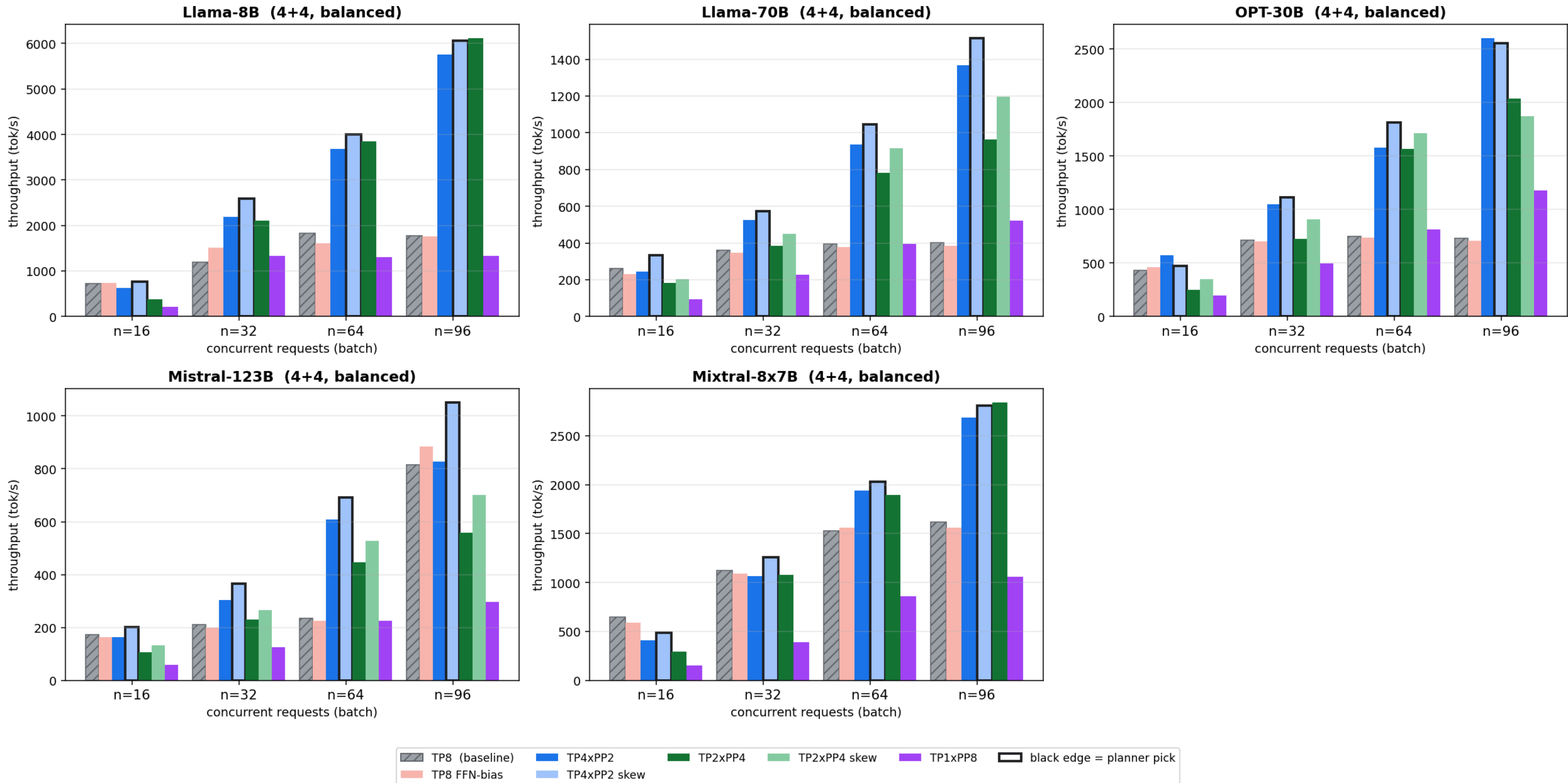
Mistral-Large-123B



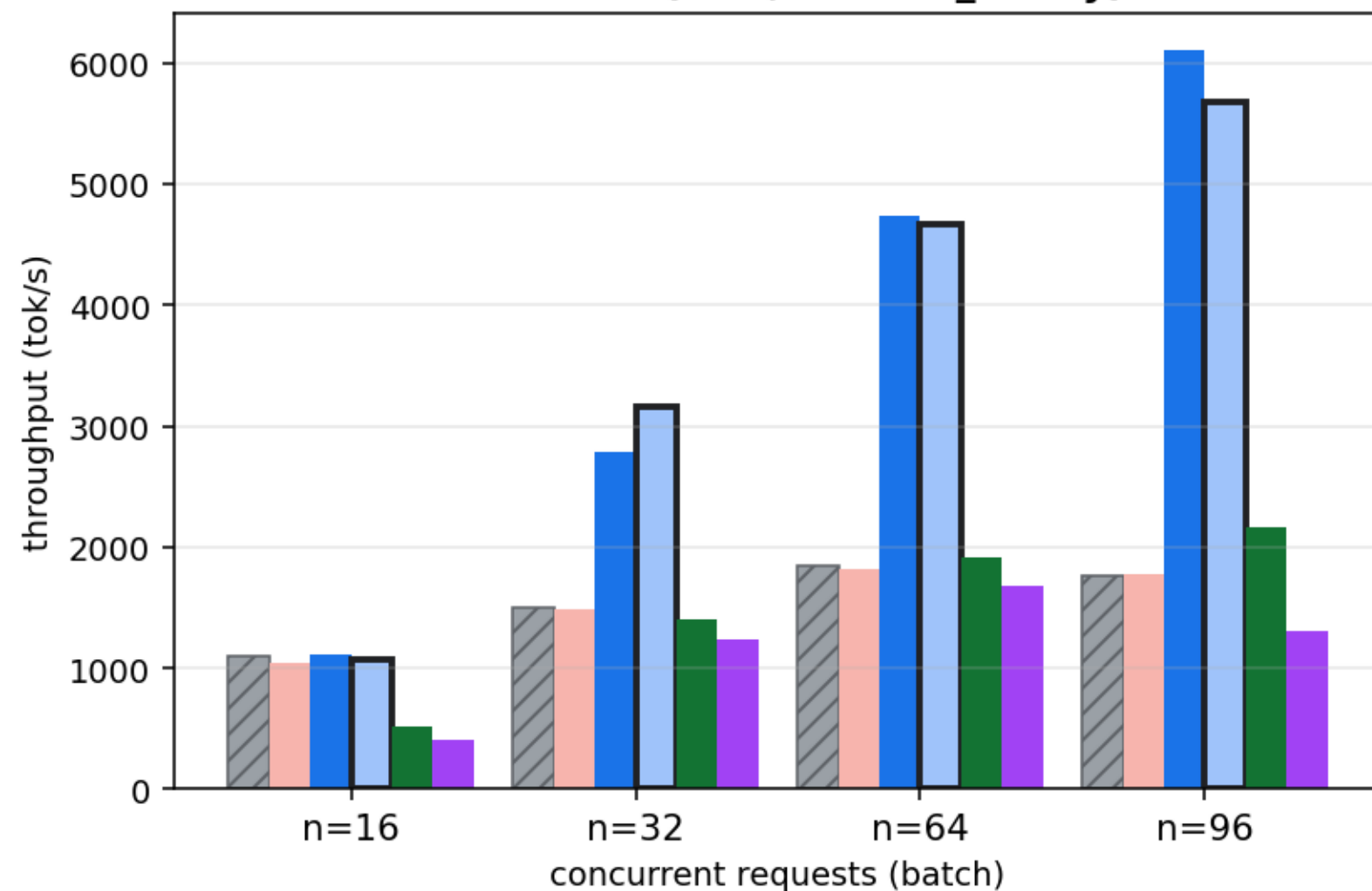
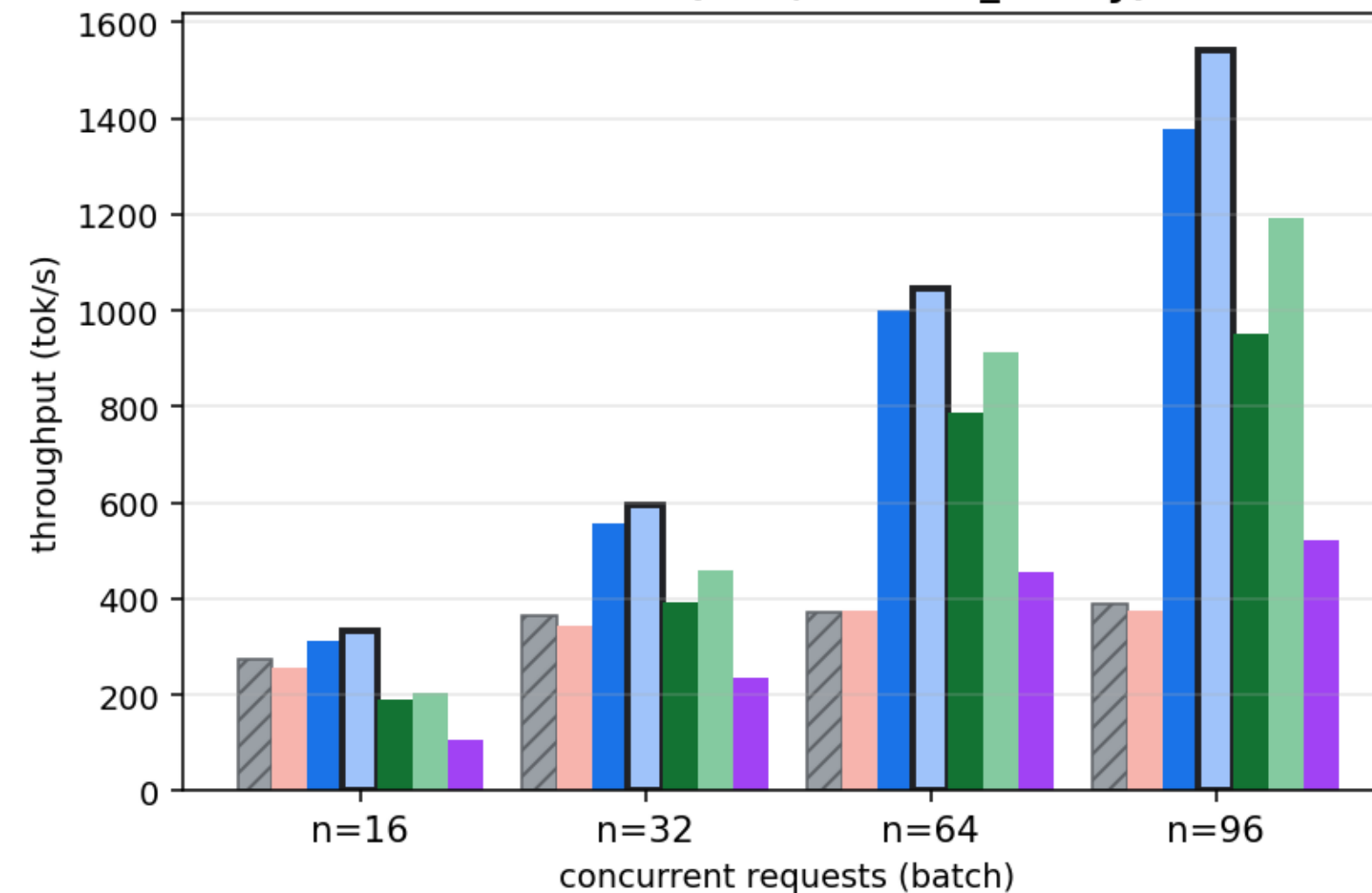
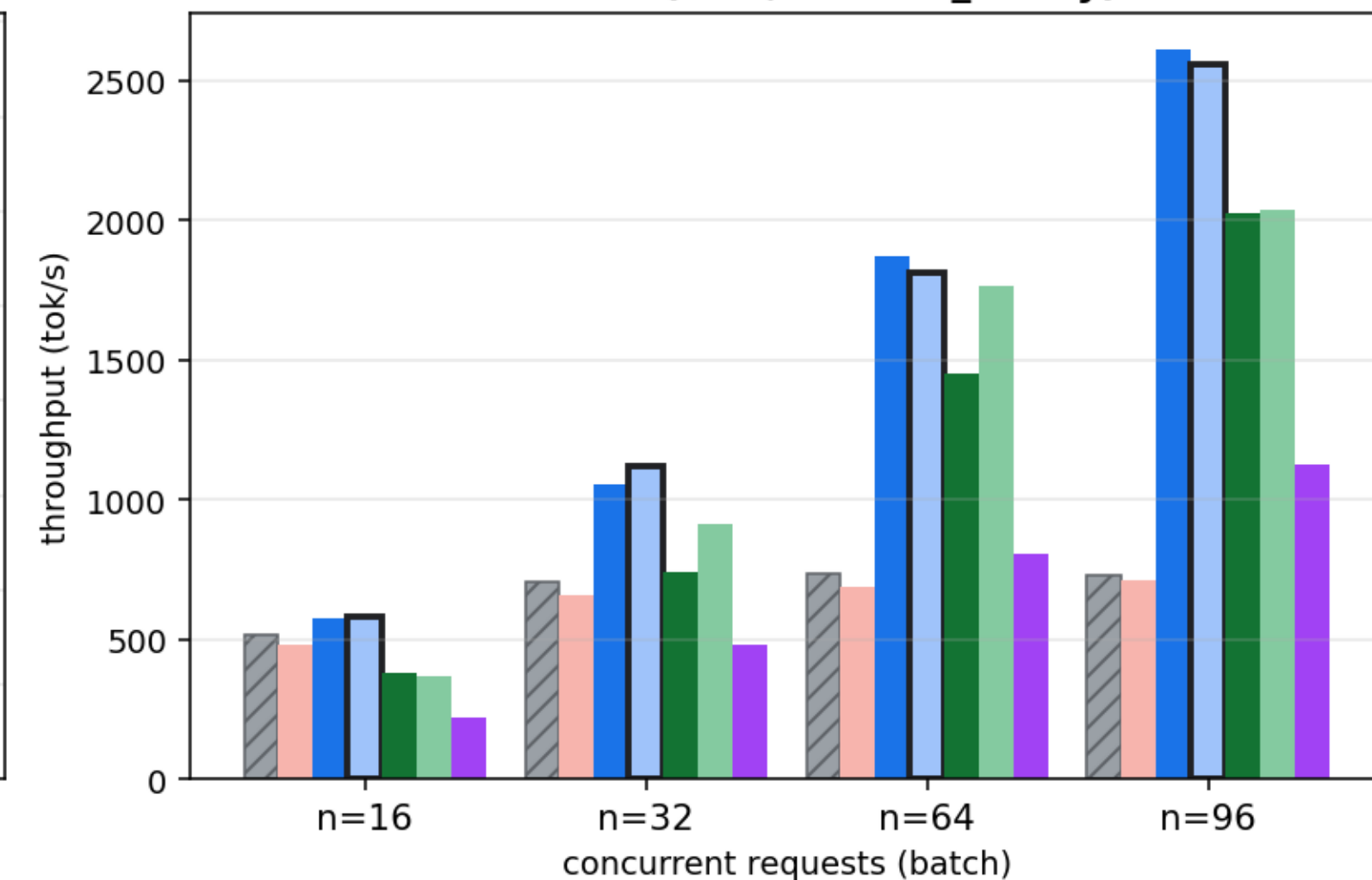
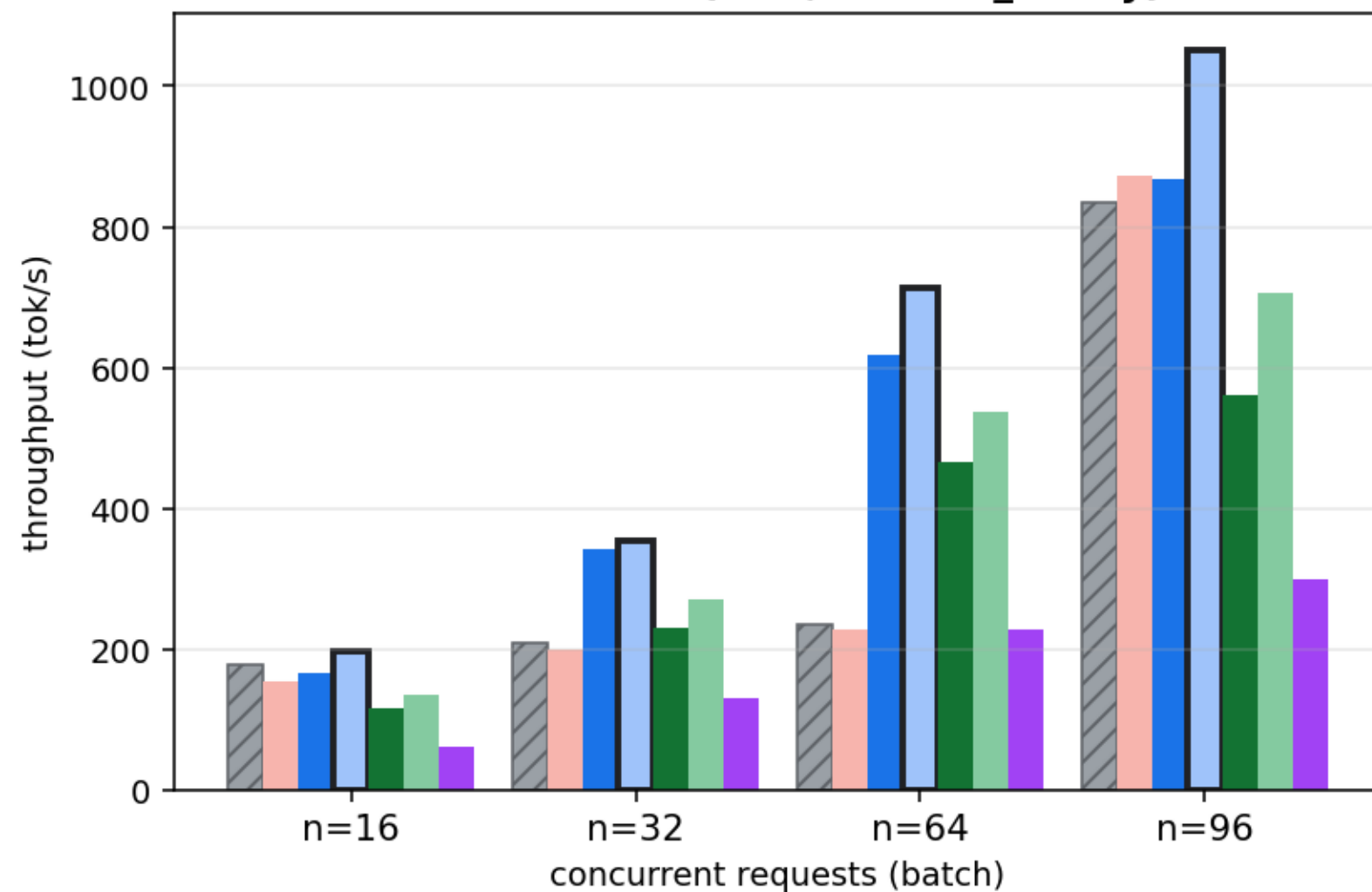
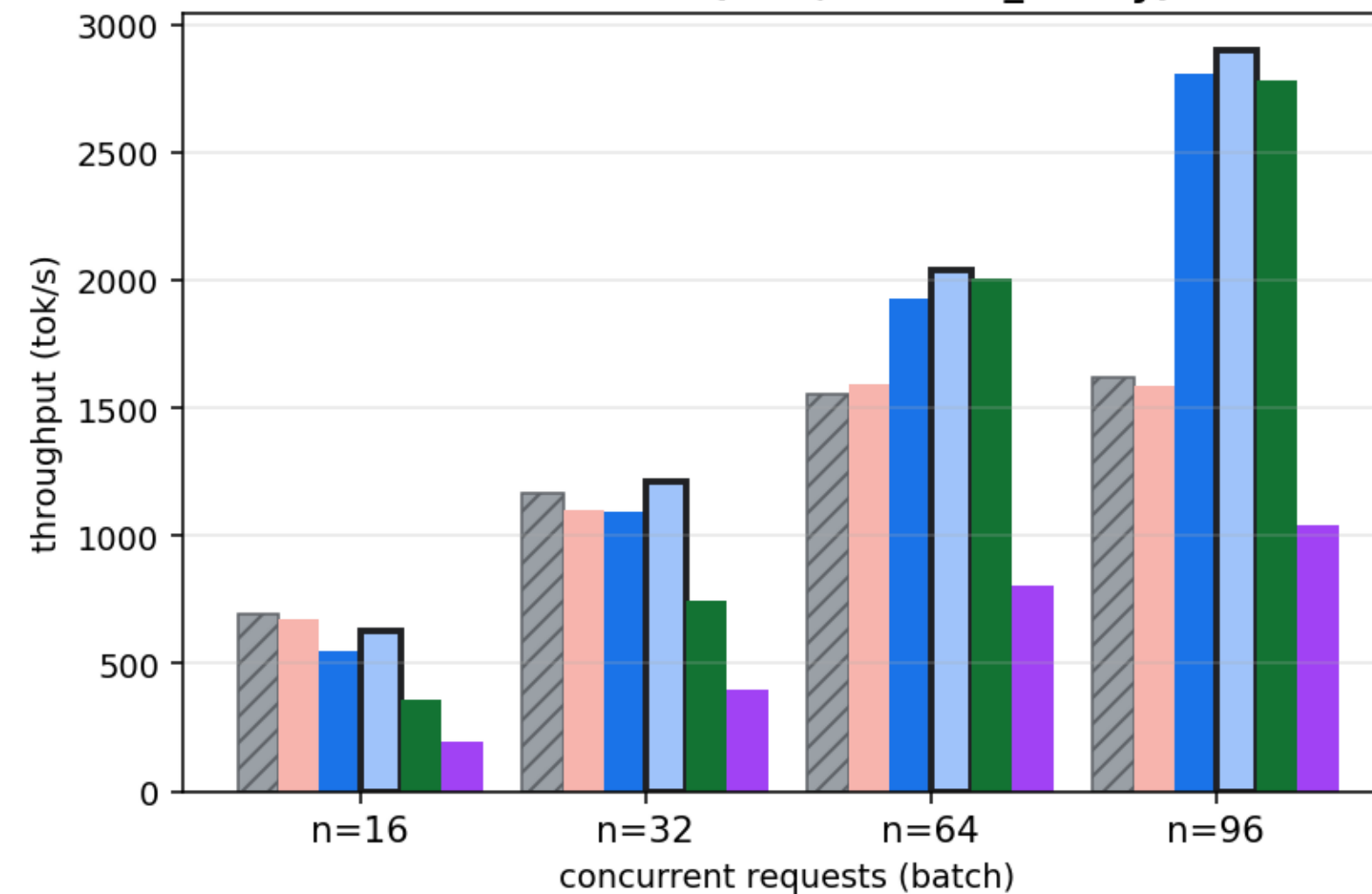
Llama-8B: which non-uniform knob matters depends on the layout
FFN-bias grows as TP shrinks; PP-skew grows as GPUs increase



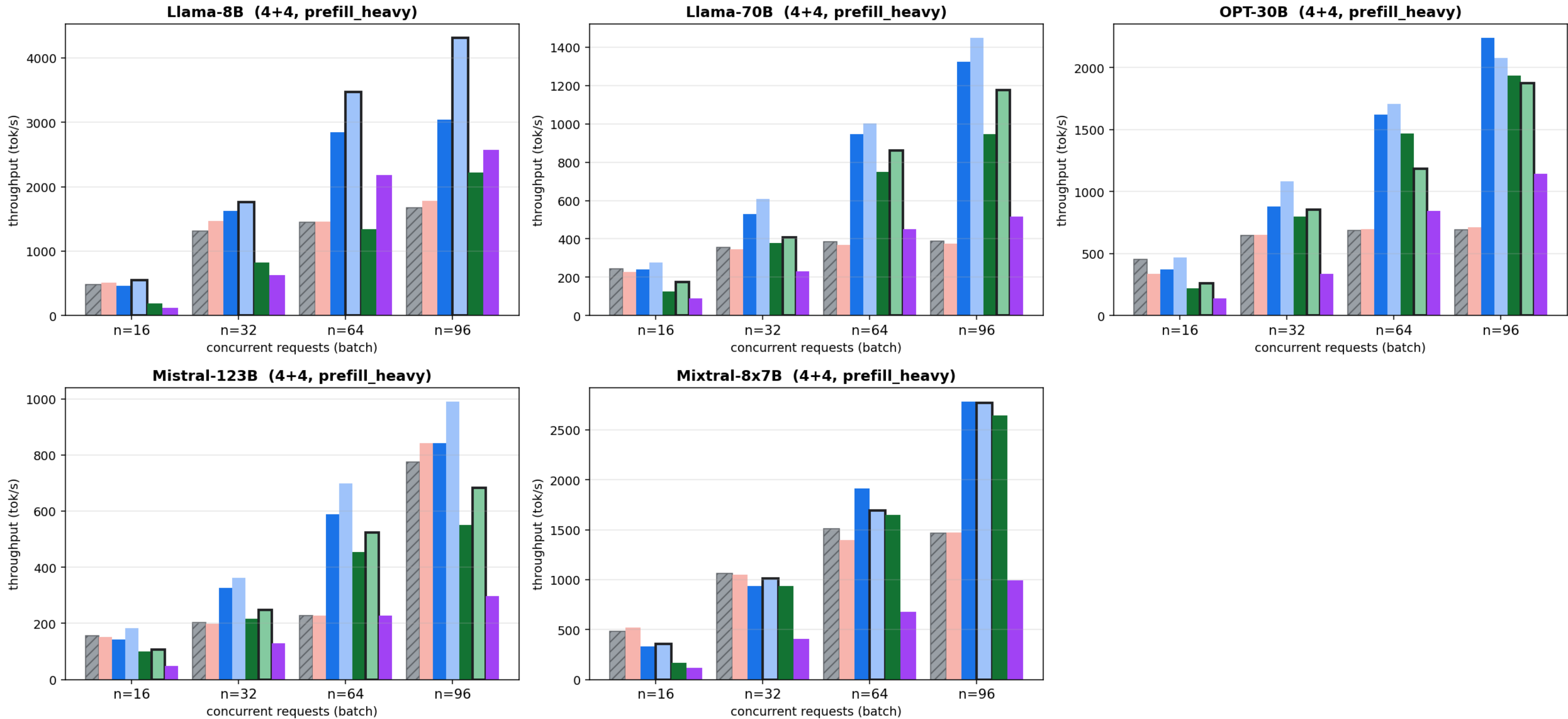
Per-model parallelization comparison — balanced, 4+4 (grouped by batch; each bar = a parallelization method)



Per-model parallelization comparison — decode_heavy, 4+4 (grouped by batch; each bar = a parallelization method)

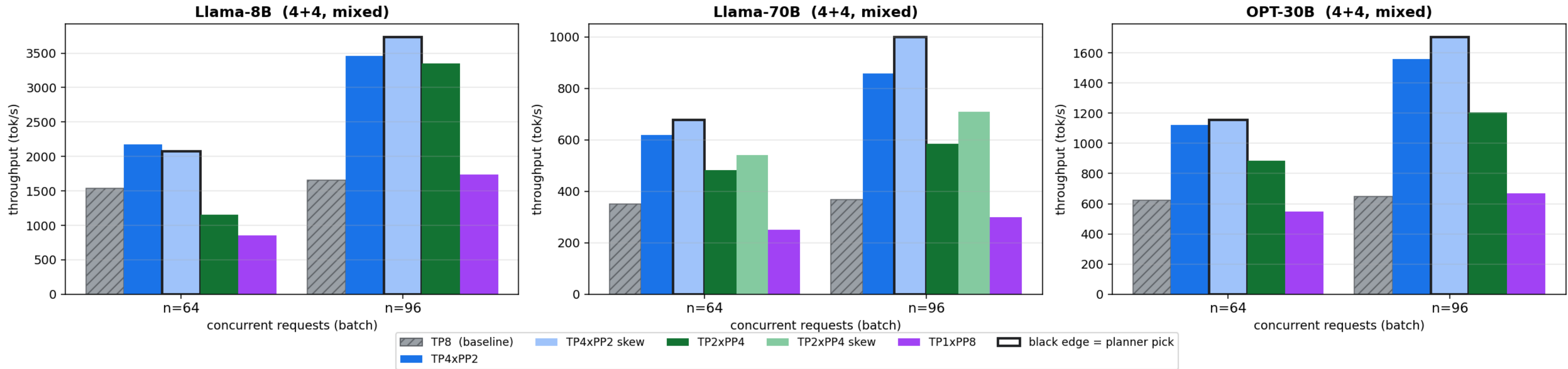
Llama-8B (4+4, decode_heavy)**Llama-70B (4+4, decode_heavy)****OPT-30B (4+4, decode_heavy)****Mistral-123B (4+4, decode_heavy)****Mixtral-8x7B (4+4, decode_heavy)**

Per-model parallelization comparison — prefill_heavy, 4+4 (grouped by batch; each bar = a parallelization method)



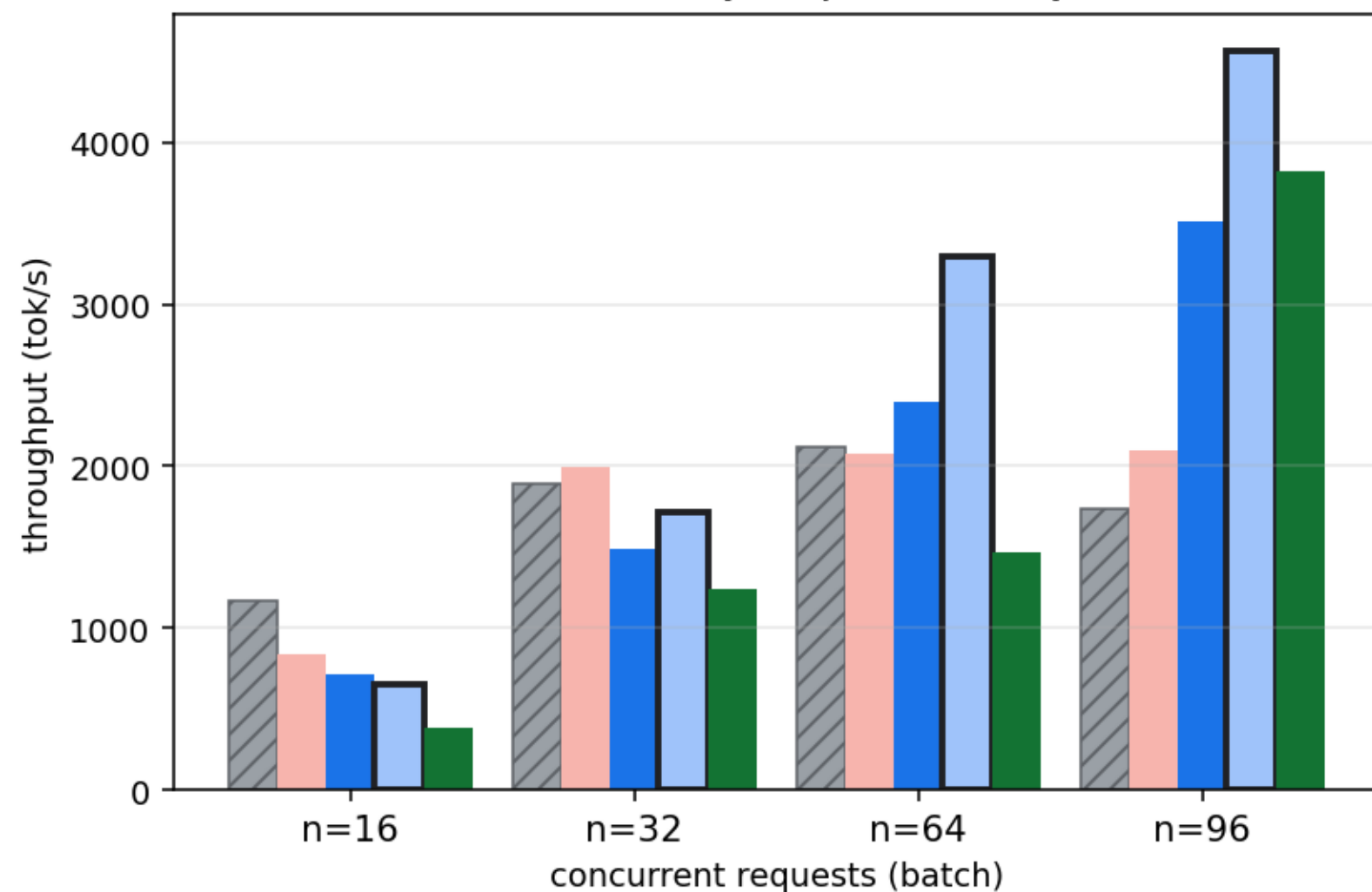
TP8 (baseline) TP8 FFN-bias TP4xPP2 TP4xPP2 skew TP2xPP4 TP2xPP4 skew TP1xPP8 black edge = planner pick

Per-model parallelization comparison — mixed, 4+4 (grouped by batch; each bar = a parallelization method)

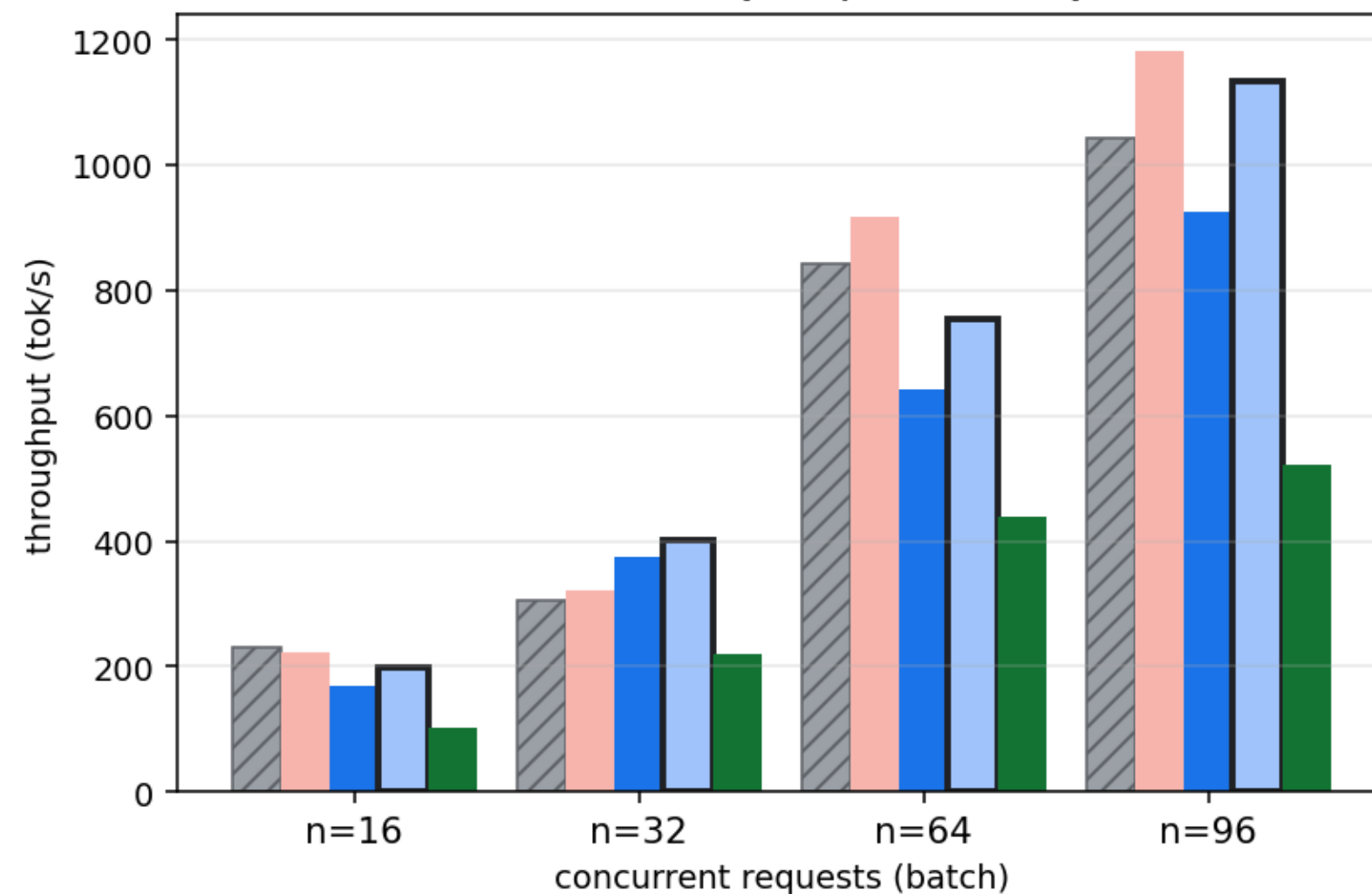


Per-model parallelization comparison — balanced, 2+2 (grouped by batch; each bar = a parallelization method)

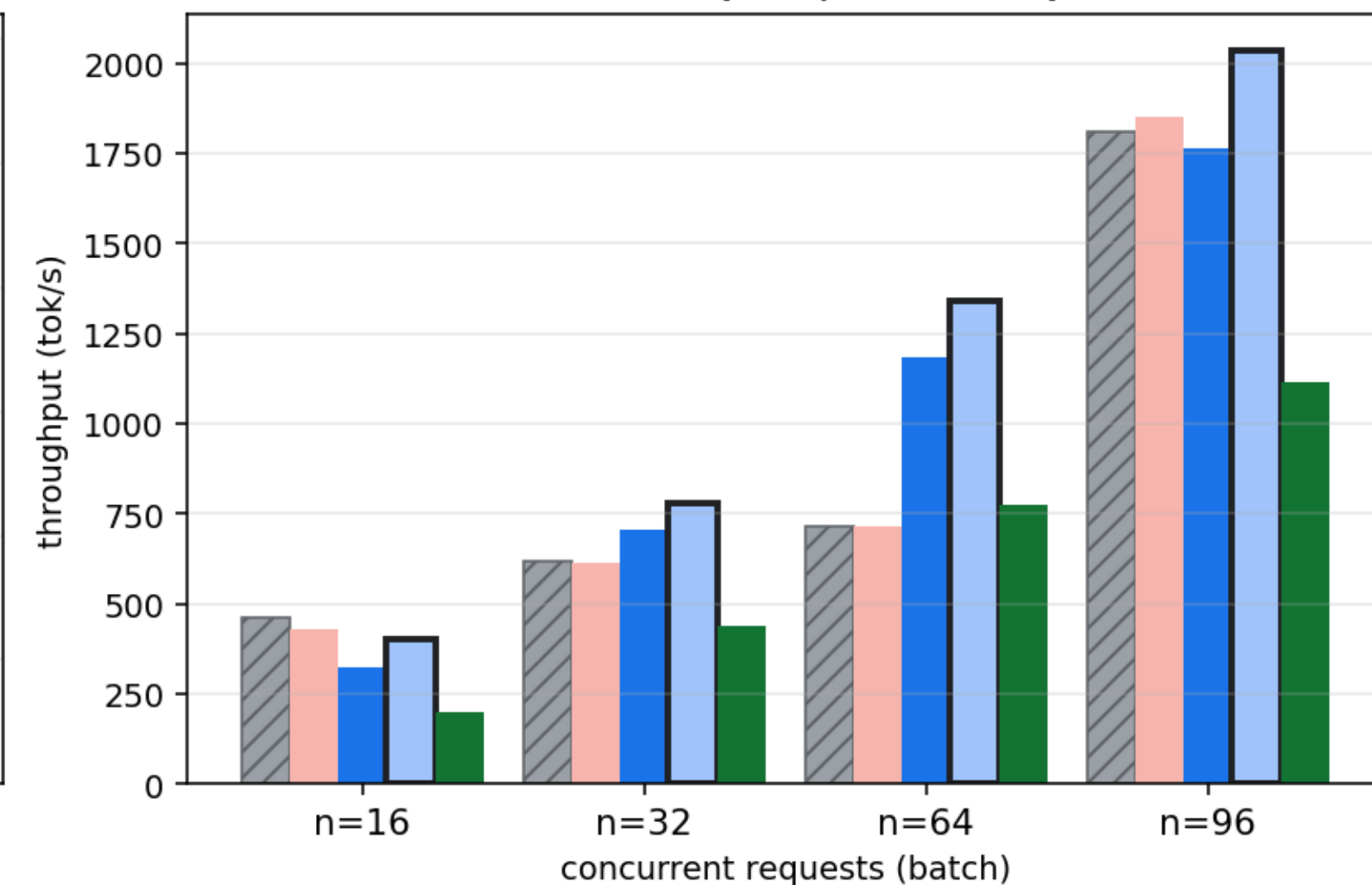
Llama-8B (2+2, balanced)



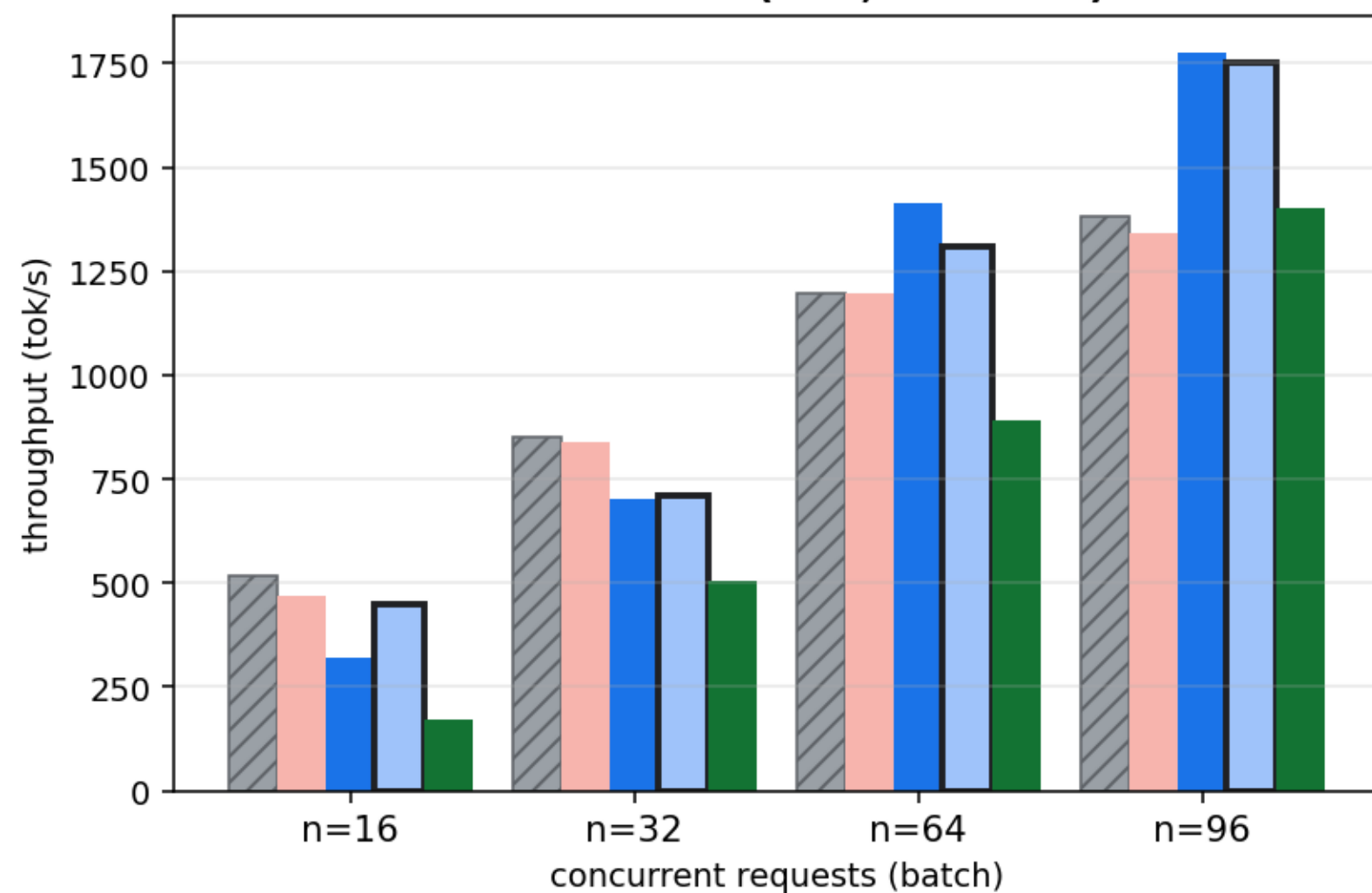
Llama-70B (2+2, balanced)



OPT-30B (2+2, balanced)

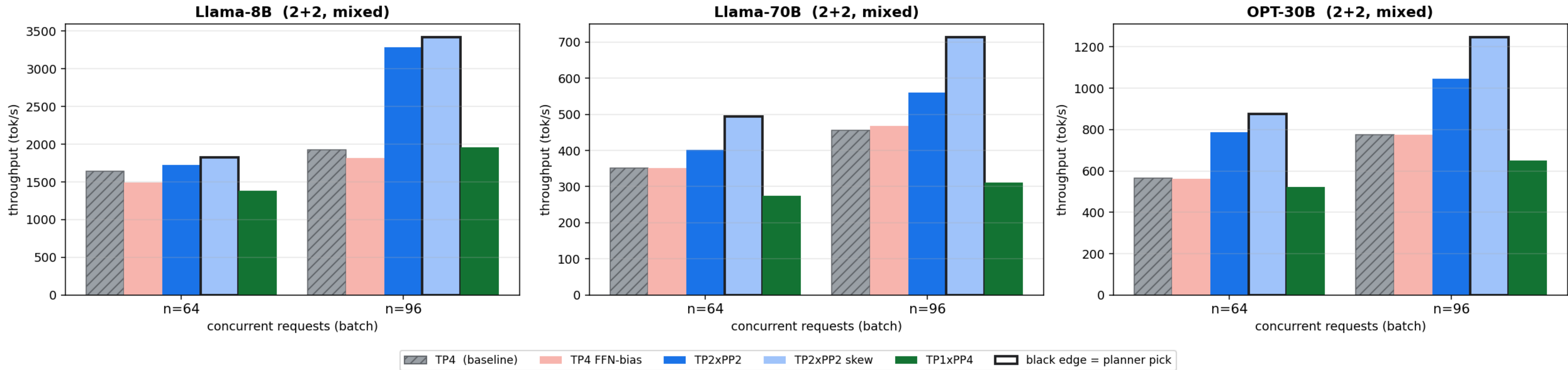


Mixtral-8x7B (2+2, balanced)



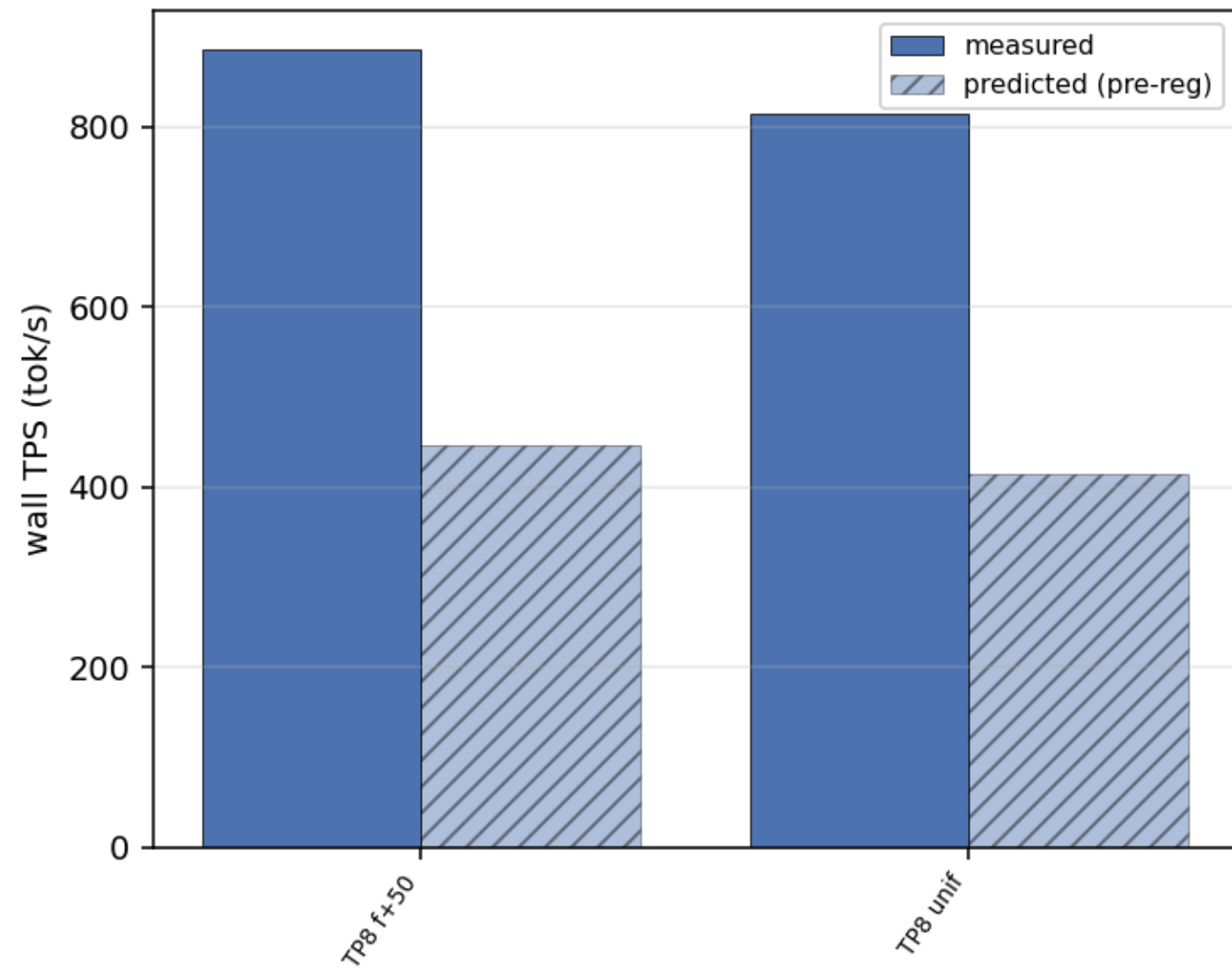
TP4 (baseline) TP4 FFN-bias TP2xPP2 TP2xPP2 skew TP1xPP4 black edge = planner pick

Per-model parallelization comparison — mixed, 2+2 (grouped by batch; each bar = a parallelization method)

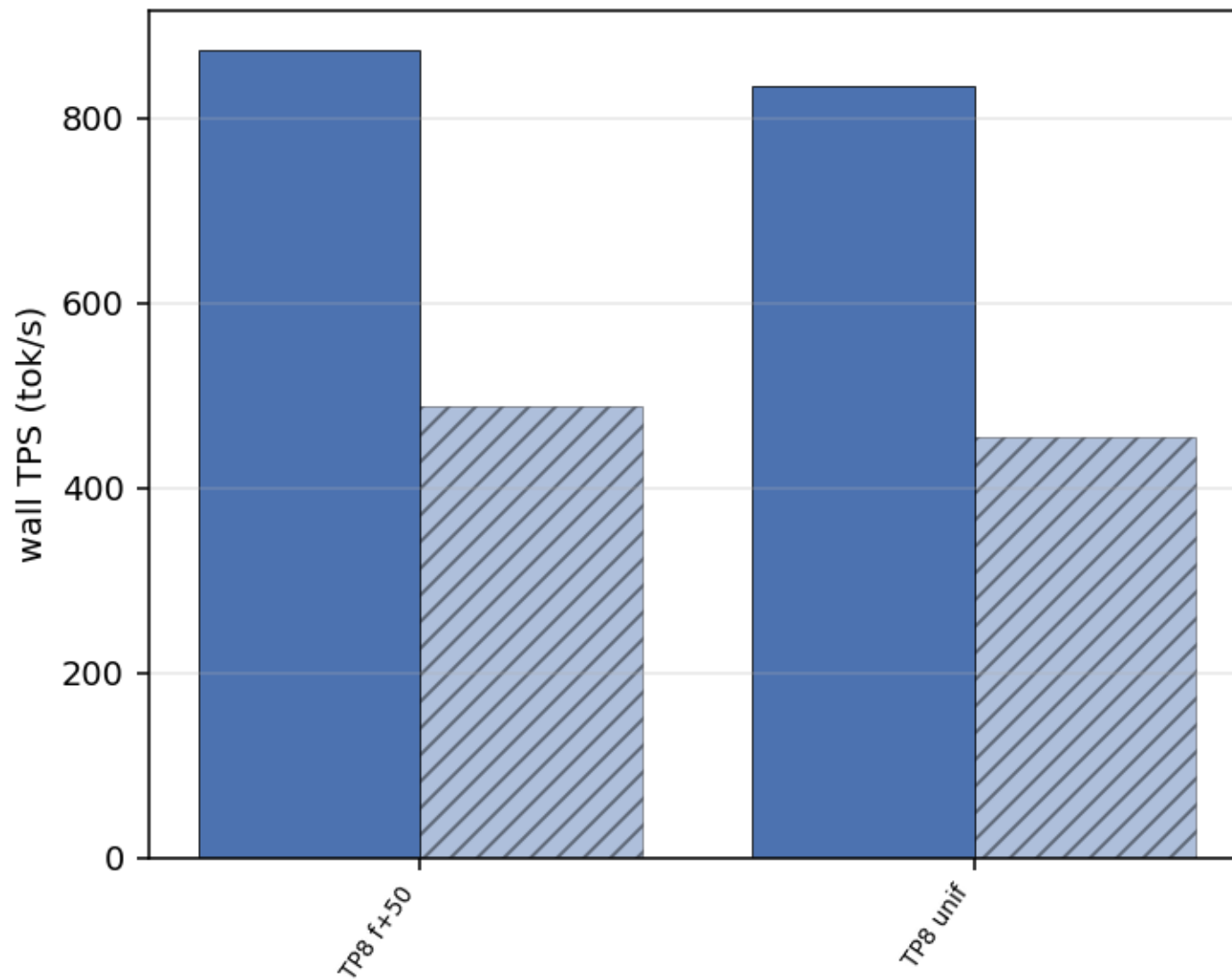


Mistral-Large-123B: PRE-REGISTERED planner prediction vs measured (frozen before sweep) — champion 3/3, mean regret 0.0%, Spearman 0.84

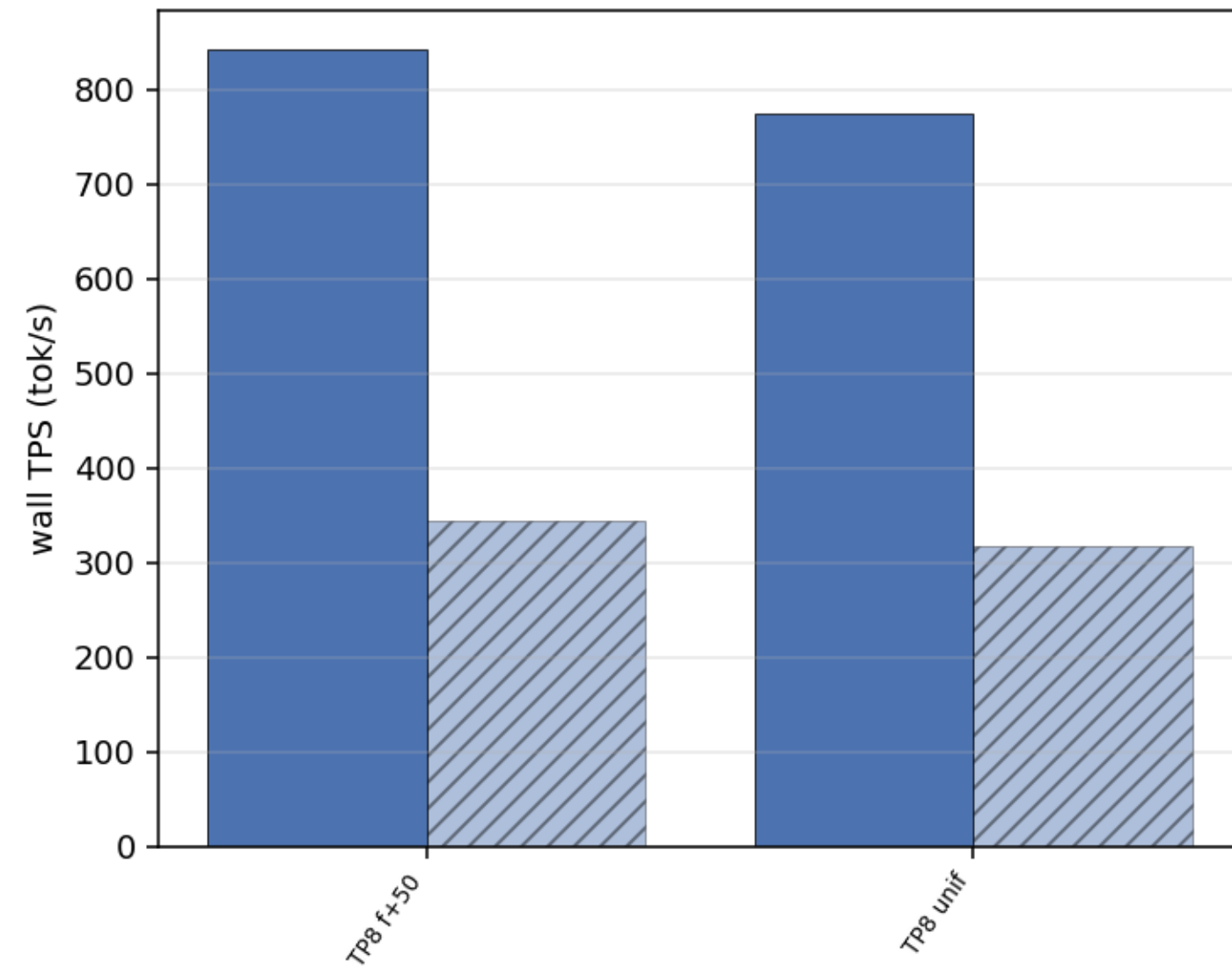
balanced
match ✓ regret 0.0% $\rho=0.75$



decode_heavy
match ✓ regret 0.0% $\rho=0.90$

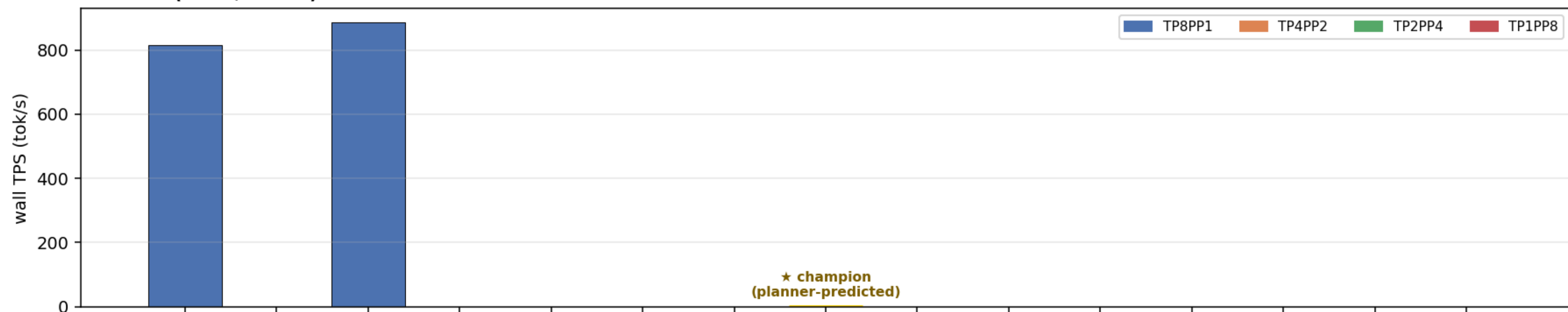


prefill_heavy
match ✓ regret 0.0% $\rho=0.85$

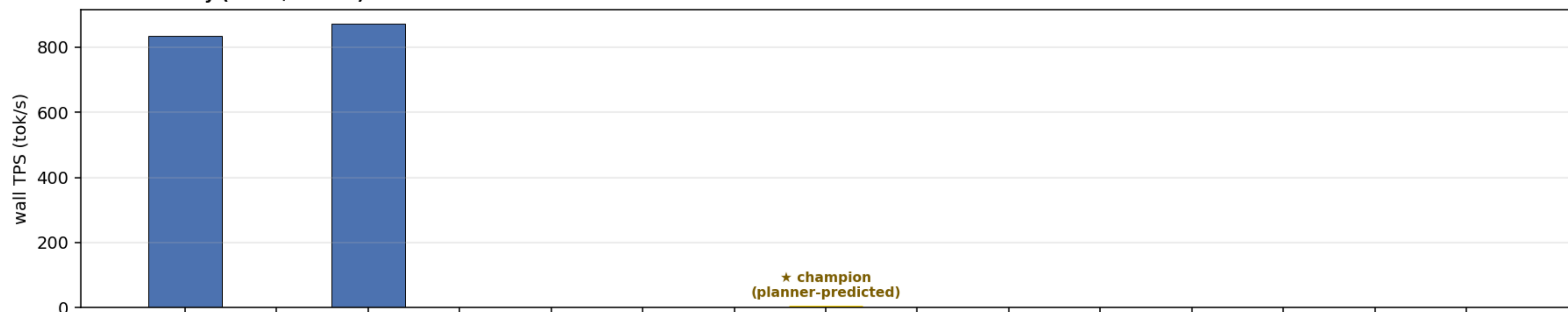


Mistral-Large-123B measured throughput — 4×Blackwell(96GB) + 4×Ada(48GB) cross-node, n_req=96

balanced (in512/out256)



decode-heavy (in128/out512)



prefill-heavy (in1024/out128)

